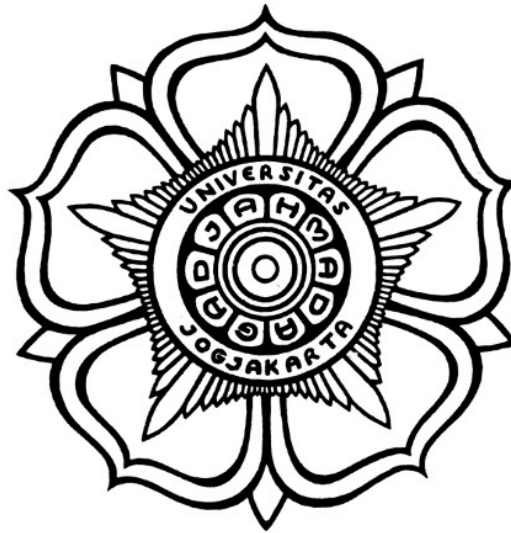


STATE FEEDBACK BASED CONTROL SYSTEM DESIGN FOR A DC-LINK COUPLED HYBRID TRAIN

SKRIPSI



THE SUSTAINABLE DEVELOPMENT GOALS
Industry, Innovation and Infrastructure
Affordable and Clean Energy
Climate Action

Written by:

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ELECTRICAL ENGINEERING PROGRAM

**DEPARTMENT OF ELECTRICAL AND INFORMATION
ENGINEERING
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YOGYAKARTA
2025**

ENDORSEMENT PAGE

STATE FEEDBACK BASED CONTROL SYSTEM DESIGN FOR A DC-LINK COUPLED HYBRID TRAIN

THESIS

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PAGE OF DEDICATION

I present this thesis to my beloved family, and my friends for their unconditional support.

PREFACE

All praise be to Allah for His mercy, grace, and guidance, which have enabled the successful completion of this thesis entitled "STATE FEEDBACK BASED CONTROL SYSTEM DESIGN FOR A DC-LINK COUPLED HYBRID TRAIN". Throughout the preparation of this final project, the author has received valuable guidance, assistance, and support from many individuals. Therefore, the author would like to express sincere gratitude to:

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NOMENCLATURE AND ABBREVIATION

ASM	= Asynchronous Motor
FOC	= Field Oriented Control
GSC	= Grid-Side Converter
LHP	= Left-Half Plane
L_{LCL}	= Inductive filter of LCL
MSC	= Motor Side Converter
PI	= Proportional-Integral
PLL	= Phase-Locked Loop
PWM	= Pulse Width Modulation
RHP	= Right-Half Plane
SG	= Synchronous Generator
$SGSC$	= Synchronous Generator-Side Converter
VOC	= Voltage Oriented Control
ω_m	= Diesel Engine Angular Velocity (rad/s)
ω_{ref}	= Angular Velocity Reference
J_e	= Diesel Engine's Inertia ($\text{kg}\cdot\text{m}^2$)
T_e	= Electromagnetic Torque ($\text{N}\cdot\text{m}$)
P_{max}	= Diesel Engine's maximum power (W)
x_{gov}	= Governor's internal status (engine speed controller)
x_{int}	= Speed controller's integrator status
K_{gov}	= Governor's gain
ω_r	= Synchronous generator's rotor angular velocity (rad/s)
J_g	= Synchronous generator's rotor inertia ($\text{kg}\cdot\text{m}^2$)
p	= Pole pair
i_d, i_q	= Synchronous generator's stator d/q-axis current (A)
v_d, v_q	= d/q-axis stator voltage (V)
L_d, L_q	= d/q-axis Synchronous inductance (H)
R_s	= Stator's resistance (Ω)
e_f	= Excitation voltage (V)
$i_{d,e}, i_{q,e}$	= Excitation's d/q-axis current
ω_e	= Excitation's angular velocity (rad/s)
$L_{d,e}, L_{q,e}$	= Excitation's inductance in d/q axis (H)
$R_{s,e}$	= Excitation stator resistance (Ω)
L_1, L_2	= LCL filter's inductance grid-side and converter-side (H)

C_f	=	LCL filter's capacitance (F)
v_c	=	LCL filters's capacitor voltage (V)
f_{res}	=	LCL filter's resonance frequency (Hz)
V_{dc}	=	DC-link voltage (V)
C_{dc}	=	DC-link capacitance(F)
R_{load}	=	DC-link resistive load (Ω)
$\text{SoC}_{1,2,3}$	=	Batteries 1 - 3's state of charge (%)
I_{bat}	=	Battery current (A)
Q_{bat}	=	Battery capacity (Ah)
$PWM_{1..4}$	=	Cell balancing PWM signal
$\text{BalEn}_{1..4}$	=	Cell balancing activation flag
K_{p1}, K_{i1}	=	Outer-loop voltage PI gain (VoC)
K_{p2}, K_{i2}	=	Inner-loop compensation voltage PI gain
K_p, K_i	=	Current loop PI gain
T_s	=	Time sampling and PWM (s)
v_{inv}	=	Inverter's reference voltage (V)
i_{ds}, i_{qs}	=	Induction motor's stator d/q current (A)
i_{dr}, i_{qr}	=	Induction motor's rotor d/q current (A)
L_{ls}, L_{lr}	=	Stator and rotor's inductance leak (H)
L_m	=	Magnetization inductance (H)
R_r	=	Induction motor's rotor resistance (Ω)
ω_{motor}	=	Mechanical angular velocity (rad/s)
T_{load}	=	Torque load (N·m)
J_m	=	Motor's Inertia ($\text{kg}\cdot\text{m}^2$)

INTISARI

Sektor per-keretaapian menghadapi tekanan yang semakin besar perihal efisiensi energi dan mengurangi emisi, yang mendorong transisi menuju teknologi lokomotif listrik hibrida (HEL). Meskipun hibridisasi menawarkan manfaat signifikan dalam penghematan bahan bakar dan fleksibilitas operasional, integrasi sistem penyimpanan baterai langsung ke DC-link menghadirkan tantangan kritis dalam manajemen aliran daya dan stabilitas sistem. Permintaan daya motor traksi yang berfluktuasi, dikombinasikan dengan inersia terbatas konverter elektronik daya, dapat menyebabkan ketidakstabilan tegangan DC-link dan peningkatan tekanan pada komponen penyimpanan energi. Meskipun penelitian sebelumnya telah secara ekstensif mengeksplorasi strategi manajemen energi dan konfigurasi perangkat keras untuk lokomotif hibrida, investigasi komprehensif terhadap metode kontrol umpan balik keadaan untuk sistem dengan kopling langsung baterai-ke-DC-link masih terbatas. Studi ini bertujuan untuk mengatasi kesenjangan ini dengan mengembangkan kerangka kerja pemodelan dan kontrol terintegrasi yang memastikan operasi yang stabil dalam kondisi beban rel kereta api yang sangat dinamis. Penelitian ini dimulai dengan pengembangan model dinamis terperinci dalam kerangka acuan dq berputar untuk generator sinkron dan motor induksi, dengan Kontrol Berorientasi Medan (FOC) yang diimplementasikan untuk mencapai kinerja torsi seperti motor DC. Model ruang-keadaan nonlinier lengkap dengan 18 variabel keadaan menangkap dinamika elektromekanis di seluruh komponen sistem, termasuk sistem governor-engine, mesin sinkron, filter LCL, DC-link, konverter dwiarah, dan motor traksi dengan FOC. Model ini dilinearisasi secara sistematis di sekitar titik operasi yang representatif untuk memfasilitasi analisis stabilitas dan desain pengontrol. Penilaian stabilitas sinyal kecil dan transien dilakukan, dengan analisis pengendalian dan observabilitas yang memverifikasi bahwa semua keadaan sistem dapat diatur dan diestimasi secara efektif. Hasil menunjukkan bahwa strategi kontrol umpan balik keadaan yang diusulkan berhasil menjaga stabilitas tegangan DC-link sekaligus mengoordinasikan aliran daya antara generator diesel dan penyimpanan baterai. Lebih lanjut, arsitektur kontrol secara efektif memitigasi osilasi tegangan dan memastikan kinerja yang tangguh di berbagai skenario operasional, memberikan fondasi untuk peningkatan efisiensi dan keandalan dalam sistem kereta api hibrida modern.

Keywords : Hybrid Electric Locomotives, DC-Link Stability, State-Feedback Control Power Flow Management, Field-Oriented Control, Battery Integration

ABSTRACT

The railway sector faces increasing pressure to improve energy efficiency and reduce emissions, driving a transition toward hybrid electric locomotive (HEL) technologies. While hybridization offers significant benefits in fuel savings and operational flexibility, the integration of battery storage systems directly onto the DC-link introduces critical challenges in power flow management and system stability. The fluctuating power demands of traction motors, combined with the finite inertia of power electronic converters, can lead to DC-link voltage instability and increased stress on energy storage components. Although previous research has extensively explored energy management strategies and hardware configurations for hybrid locomotives, a comprehensive investigation into state-feedback control methods for systems with direct battery-to-DC-link coupling remains limited. This study aims to address this gap by developing an integrated modeling and control framework that ensures stable operation under highly dynamic railway loading conditions. The research begins with the development of detailed dynamic models in the rotating dq-reference frame for both the synchronous generator and induction motor, with Field-Oriented Control (FOC) implemented to achieve DC motor-like torque performance. A complete nonlinear state-space model with 26 state variables captures the electromechanical dynamics across all system components, including the governor-engine system, synchronous machine, LCL filter, DC-link, bidirectional converter, and traction motor with FOC. The model is systematically linearized around representative operating points to facilitate stability analysis and controller design. Both small-signal and transient stability assessments are conducted, with controllability and observability analyses verifying that all system states can be effectively regulated and estimated. The results demonstrate that the proposed state-feedback control strategy successfully maintains DC-link voltage stability while coordinating power flow between the diesel generator and battery storage. Furthermore, the control architecture effectively mitigates voltage oscillations and ensures robust performance across various operational scenarios, providing a foundation for enhanced efficiency and reliability in modern hybrid railway systems

Keywords : Hybrid Electric Locomotives, DC-Link Stability, State-Feedback Control Power Flow Management, Field-Oriented Control, Battery Integration

CHAPTER I

INTRODUCTION

1.1 Research Background

The railway sector is under increasing pressure to improve energy efficiency and reduce emissions. Conventional diesel locomotives, while reliable, face challenges in meeting modern environmental standards and efficiency demands. Hybrid electric locomotive (HEL) technology has emerged as a promising solution, combining conventional diesel power with electric energy storage systems to enhance fuel economy, reduce emissions, and improve operational flexibility [4].

One of the main technical challenges in HEL systems lies in the management of power flows between the diesel generator, the traction motors, and the energy storage system. The integration of batteries directly coupled to the DC link provides opportunities for regenerative braking capture and load leveling but introduces difficulties in maintaining DC-link stability and controlling battery current within safe limits [1]. Proper control of these interactions is essential to ensure reliable performance and extend the lifetime of the energy storage system.

Another critical aspect is power electronic conversion and control. Studies on advanced converter topologies and modulation strategies demonstrate that the choice of switching and control methods directly affects system efficiency, current quality, and ripple on the battery side. For example, optimized PWM strategies and advanced current control techniques can significantly reduce harmonic distortion and current ripple, thereby improving overall system stability and reducing stress on the storage elements [2]. These findings indicate the importance of carefully designed control loops that integrate both AC-side and DC-side requirements.

At the same time, energy management strategies at the system level play a vital role in hybrid locomotive design. Frequency-based power-split methods, for example, allocate low-frequency power demands to the diesel generator and high-frequency fluctuations to storage devices, thereby minimizing battery stress while maintaining high efficiency [3]. Such approaches highlight the need for coordination between supervisory energy management and lower-level control systems.

The literature consistently reports substantial benefits of hybridization in locomotives, including reductions in fuel consumption, lower emissions, and noise attenuation in sensitive areas [4]. However, while energy management strategies and hardware implementations have been widely explored, there remains a lack of research focusing specifically on state-feedback control methods for hybrid railway systems where the battery is

directly coupled to the DC link. Addressing this gap is essential to ensure stable operation under varying load conditions and to mitigate the negative impacts of ripple currents on battery health.

Based on these challenges, this research proposes the design and analysis of a state-feedback control system tailored to hybrid locomotives with a direct battery-to-DC-link connection. By integrating insights from system architecture, and power electronics, the proposed approach aims to stabilize the DC link, regulate battery currents, and support the efficiency and durability goals of modern hybrid rail systems.

1.2 Problem Identification

Based on the background study, the core problems of this thesis can be addressed as follows:

1. How can batteries be coupled into the DC link to share power effectively with the diesel generator?
2. How can the DC link voltage be stabilized under highly dynamic load conditions?
3. How can power electronic converters and advanced control strategies ensure seamless power flow between multiple sources and loads?
4. How can a state feedback control system be designed to provide both stability and performance in real-life railway applications?

1.3 Aim of the Study

The primary purpose of this research is to conduct a comprehensive stability assessment and analysis of the relationship of power stability between converters and the Synchronous Machines (SM), focusing on the dynamic control architectures. This thesis aims to systematically investigate the dynamic behavior and identify the stability limits of these converter under a wide range of operational conditions using both small-signal and transient analysis. To achieve this, the following specific objectives are established:

1. Address the challenge of fluctuating traction power demand by utilizing a combination of diesel generator and battery support instead of oversizing the diesel engine.
2. Develop power electronic control methods that coordinate power flow seamlessly between the diesel generator, rectifier, battery, and traction inverter.
3. Achieve stable DC link voltage under dynamic load variations through advanced control strategies.
4. This study emphasizes the design of state feedback control.

5. Implement a state feedback control system with the goal of enhancing the quality of the real-life railway applications.

1.4 Research Limitations

This thesis has several limitations that affect the scope and the results of the study, which are as follows:

1. The study is limited to a simulation-based analysis using MATLAB/Simulink, with no experimental prototype being carried out.
2. The synchronous generator are modeled to be a dynamic dq frame, focusing on their interactions with the converters and DC-Link.
3. Both PWM rectifier and inverter are assumed to be ideal switching devices, neglecting other losses.
4. The batteries are assumed to be lumped and the parameters for the state-of-charge (SoC), voltage, current, without detailed electrochemical aging and thermal effects.
5. Grid connection, regenerative braking recovery to external networks, and other infrastructure-level interactions are not included in the scope.
6. The results are validated through simulation scenarios under predetermined load conditions and parameters, with no experimental operations testing.

1.5 Benefit of Research

1. Contributes to the development of advanced control strategies (state feedback in dq frame) that enhance DC link stability, improve system robustness, and ensure efficient power flow.
2. Identify the impact of DC-Link's stability in reducing the voltage ripple and oscillations.
3. Provides a systematic approach to stabilizing the load-flow and etc. using the DC-Link as the central energy exchange point between diesel generator, battery, and traction motor.

1.6 Systematic of Writing

The structure of this thesis is organized by presenting the background, objectives, methodology, analysis, scenarios, and conclusions. The systematic writing of this thesis is as follows:

1. **Introduction:** This chapter contains the background of the research, problem identification, research objectives, research limitations, benefits of the research.

2. **Basic Theory and Literature Review:** This chapter discusses the theoretical foundations and previous research related to the topic. It includes relevant theories, concepts, and frameworks that support the research.
3. **Research Methodology:** This chapter outlines the research design, data sources, data collection techniques, and data analysis methods used in the study.
4. **Analysis and Discussion:** This chapter presents the analysis of the data collected during the research.
5. **Scenarios:** This chapter explores various scenarios related to the research findings.
6. **Conclusion and Recommendations:** This chapter summarizes the key findings of the research and provides conclusions based on the analysis and offers recommendations for future research.

CHAPTER II

LITERATURE REVIEW AND FUNDAMENTAL THEORY

2.1 Literature Review

This section presents a comprehensive review on the control design and analysis of the Hybrid Powertrain System for railway applications. The various control of the system relies on its control algorithms to achieve the desired output characteristics. A hybrid system typically combines multiple controllers that are divided into three primary sections: the power synchronization control, the voltage-current control loops, and lastly the energy management system (EMS) that oversees the overall operation of the hybrid system. The Power synchronization control is responsible for regulating the active and reactive power flow produced by the rectifier and the inverter for their respective machine in the Point of Common Coupling (PCC) perspective. While many algorithms exist for power synchronization in a powertrain system, one of the most prominent methods are a double closed loop control with the usage of space vector with a focus on the voltage and the current to modulate the PWM switching of the rectifier and the inverter [1].

2.1.1 Converter

In the research conducted by writer Qian Kailiang, titled *Comparative Study of some Control Techniques Applied to Three-Phase PFC Rectifiers*. This journal demonstrates the comparison of two control methods for a three-phase power factor correction (PFC) rectifier. The primary comparison revolves around the control techniques used in a three-phase stationary coordinate system and in a two-phase rotating system [1]. The two methods are the Voltage Oriented Control (VoC) and the Direct Power Control (DPC). The VoC method utilizes a double closed loop control structure with the outer loop regulating the DC link voltage and the inner loop regulating the input current. The DPC method, on the other hand, directly controls the active and reactive power by selecting appropriate voltage vectors based on the instantaneous power errors. The study concludes that while both methods are effective [1]. These control methods are compared in terms of their dynamic response, harmonic distortion, power factor correction, and energy efficiency. Results show that the method using VoC achieves a better and lower harmonic performance while giving a higher power factor, but with the drawback of a slower dynamic response in its dependence on the PI controller tuning and its sensitivity to parameter mismatches. While DPC offers a faster dynamic response and robustness, but it suffers from higher harmonic distortion and generally lower power factor compared to VoC [2].

Researcher Lyu S, Zheng L titled *A Start-Up Inrush Current Suppression Method*

Based on Dual Parameters Soft-start for PWM Rectifier suggests a method that suppresses the inrush current into the PWM rectifier during the start-up. This method utilizes a dual-parameter soft-start that encompasses soft-start modulation ratio and soft-start DC voltage reference. This journal focuses on the analysis of the spike mechanism and obtaining the mathematical model from the rectifier. The method presented effectively diminish the current spike [3].

At most cases, a PWM rectifier is typically paired with a load inverter as reference from the journal titled *Analysis and Control of Three Phase PWM Rectifier for Power Factor Improvement of IM Drive* created by Ajesh P, Jisha Kuruvilam, Dr. Anasraj can be referred for the usage of the PWM rectifier in a train system. Adding to the previous research, the journal titled *Subway Traction Power Supply System Based on PWM Rectifier* by Liu X shows that power factor increases in proportion to the load that is connected to the inverter with the changes in torque of the Induction Motor (IM). This causes the increase in the requirement of the current in the inverter thus the control regulation of the PWM rectifier [4].

2.1.2 Synchronous and Asynchronous Machine

In the larger scheme of the hybrid powertrain system, modern hybrid locomotive system often integrates the AC to DC rectifier as parts of a cascade propulsion involving alternators, traction inverters, energy storage devices and mechanical drivetrain components. In accordance to the journal titled *Hybrid Electric locomotive Powertrains* by Hou explains that hybrid locomotives typically adopt a series diesel-electric architecture where a synchronous generator driven by a diesel engine produces electrical power that is subsequently converted by the rectifier and supplies the DC-Link which feeds the traction inverters and induction motors [5]. Improvement in power electronics have enabled the transition from conventional DC machines to AC synchronous and induction machines, that yields higher efficiency, improved reliability, and a reduced maintenance requirements.

Acting as the main source of power in the hybrid system, the synchronous generator converts mechanical input into electrical. The generator's field excitation plays a major role in regulating the terminal voltage and maintaining system stability under fluctuating traction loads. Earlier works such as *Motor-Alternator Speed Regulator for the New Hacen Speed-Merchant Locomotive*, highlight the primary constraints such as voltage build up behaviour, transient load rejection, and maintaining constant voltage and frequency under sudden input disturbances. While historic locomotive systems relied on magnetic amplifier-based regulators, modern system employ a high-bandwidth automatic voltage regulator (AVR) and electronic excitation controllers that integrates with real-time feedback of the generator's current, field voltage, and rotor speed. These

sub-systems ensures that the generator maintains steady output even when the traction inverters impose rapid load changes that associates with torque transients [6].

As the final component in the system. In accordance to the journal of *Hybrid Electric Locomotive Powertrains*, traction motor typically operate in a three to five distinct regions: maximum torque at low speed, maximum power at intermediate speeds, constant speed cruising, coasting, and regenerative braking. Each region imposes different current, voltage, and power requirements on the inverter. Vector control or Field Oriented Control (FoC) is the most common method applied to regulate the motor's torque and flux, and high-frequency PWM switching strategies is used to ensure smooth torque production with minimal ripple. In a real life situation of adhesion loss, the inverter must rapidly reduce the motor's torque to prevent any wheel slip while still maintaining stability of the mechanical to electrical interaction [5]. These situation plays a crucial role in the traction performance of the locomotive. In the *Functional Efficiency Enhancement of Diesel-Electric Locomotive Traction System* by Vira N. demonstrate that the integration of advanced control strategies such as torque and current control independently via rectifier and inverter, the system rapidly detects and corrects any slip in the induction machine [7]. In this research, two different methods of control are used, one focused on the generator-to-rectifier dynamics while the other focuses on the dynamics between the source and the load for which in this case is the Induction Motor (IM). Although the use of multiple control methods allows improved transient performance and tighter regulation of the DC-link and torque, it also introduces potential interaction and coordination challenges between control loops. Therefore, this work evaluates both local-loop designs (generator/rectifier and rectifier/inverter-IM) and supervisory coordination strategies such as feedforward decoupling, gain scheduling, and an energy management layer to ensure stability and optimal power sharing under varying loads and slip events. Performance is assessed by metrics including DC-link voltage deviation, torque tracking, harmonic distortion, and robustness to parameter variations [8].

2.2 Fundamental Theory

2.2.1 PWM Rectifier

A passive diode rectifier and thyristor rectifier which its phase angle are controlled can be commonly find in most of powertrain systems. Passive rectifier brings a myriad of advantages such as its simplicity in control algorithm [9], but this type still has its drawback by its inability to handle the harmonic distortion that effects the system significantly and the primary drawback which is the existance of a lagging element that occurs at the input ultimately reducing the efficiency of the system along with the harmonic distortion increase of the current in the AC input [10].

To overcome these drawbacks with the wishes of being able to create a well reg-

ulated larger voltage output with little to zero ripple, modern powertrain system often employs a PWM rectifier. A PWM rectifier is a type of rectifier that in theory, the semi-conductor switches are to be controlled via PWM (Pulse Width Modulation) and the switching frequency is to be much higher than the fundamental frequency of the AC input. By controlling the switching of the semi-conductor switches, the output voltage can be regulated to a desired value that in most cases are always higher than the input voltage [10].

Among the many usage of the PWM rectifier, its advantages can be summarized as follows:

- **Bidirectional Power Flow:** In the realms of electric power generation and distribution and electric vehicles (such as microturbine, wind turbine for generation, and a crane, elevator, and most cases of electric vehicle), the ability to facilitate bidirectional power flow is crucial. PWM rectifiers can seamlessly convert AC to DC during power generation and reverse the process during regenerative braking, allowing energy to be fed back into the grid or stored in batteries [10].
- **Sinusoidal Input Current:** Input current from the PWM rectifier that are connected to the grid has the form of a sinusoidal with a low to middle harmonic distortion that occurs at the carrier frequency of the switching. Though this harmonic disturbance can be further reduced with the usage of filters such as LCL filter, the PWM rectifier inherently produces a cleaner sinusoidal current waveform compared to traditional diode or thyristor rectifiers [11] [12].
- **Controllable Power Factor:** The ability to control the power factor at the AC input is a significant advantage of PWM rectifiers. By adjusting the phase angle between the input voltage and current, the rectifier can operate at unity power factor or even provide reactive power support to the grid, enhancing overall system efficiency and stability [12].
- **Regulated Output Voltage:** PWM rectifiers can maintain a stable and regulated DC output voltage despite variations in the AC input voltage or load conditions. This is particularly important in applications where a consistent DC voltage is required for downstream components, such as in electric vehicle powertrains or renewable energy systems [11].

In figure [], a PWM boost rectifier topology is shown. The PWM rectifier consists of six IGBTs (Insulated Gate Bipolar Transistor) that are arranged in a three-phase bridge configuration. Each phase of the AC input is connected to a pair of IGBTs, allowing for

controlled switching to regulate the output voltage and current. To suppress the negative effects from the active rectifier, comes the need for a higher and steeper level of control to the output of the rectifier via pulse width modulation (PWM) technique. By adjusting the duty cycle of the PWM signals that control the switching of the IGBTs, the output voltage can be regulated to a desired level while maintaining a sinusoidal input current with low harmonic distortion.

2.2.1.1 Mathematical Model of PWM Rectifier in the dq -Frame

A PWM rectifier is generally modeled as a voltage-source converter (VSC) connected to the AC supply through an interface inductance L_f and a resistance R_f . When transformed to the dq -frame aligned using a phase-locked loop (PLL), the control system can independently regulate the active and reactive components of current. This forms the foundation of voltage-oriented control (VoC), widely adopted in traction and power electronic systems.

The modeling begins by converting the three-phase quantities (a, b, c) into the rotating dq -coordinates. This transformation decomposes AC variables into direct-axis (d), quadrature-axis (q), and zero-sequence components. By choosing θ to be aligned with the grid (or generator) voltage, the q -axis voltage becomes zero ($v_{s,q} = 0$), which simplifies active power control.

The mathematical form of the Park transformation is:

$$\begin{bmatrix} x_d \\ x_q \\ x_0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos \theta & \cos(\theta - \frac{2\pi}{3}) & \cos(\theta + \frac{2\pi}{3}) \\ -\sin \theta & -\sin(\theta - \frac{2\pi}{3}) & -\sin(\theta + \frac{2\pi}{3}) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix}. \quad (2-1)$$

Through this transformation, sinusoidal waveforms are mapped into constant values when steady-state conditions are reached. This greatly simplifies the derivation of dynamic equations and allows control loops to operate on DC-like signals rather than oscillatory AC waveforms.

To derive the electrical behavior of the rectifier, Kirchhoff's Voltage Law (KVL) is applied to the filter connecting the AC source and the converter. In the dq -frame, the voltages and currents become:

$$v_{s,d} = v_{conv,d} + R_f i_d + L_f \frac{di_d}{dt} - \omega L_f i_q, \quad (2-2)$$

$$v_{s,q} = v_{conv,q} + R_f i_q + L_f \frac{di_q}{dt} + \omega L_f i_d. \quad (2-3)$$

The terms $\omega L_f i_q$ and $-\omega L_f i_d$ represent the cross-coupling effects caused by the rotating reference frame. These terms illustrate how changes in the d -axis current can affect the q -axis voltage and vice versa. In practical implementations, these coupling terms are compensated using feedforward decoupling to improve the dynamic performance of the current controllers.

The converter itself is modeled using an average representation of the switching actions. Instead of directly considering high-frequency PWM behavior, an averaged voltage is used to represent the fundamental component of the PWM output. For a two-level VSC, the dq-frame converter voltage becomes:

$$v_{conv,d} = \frac{V_{dc}}{2} m_d, \quad v_{conv,q} = \frac{V_{dc}}{2} m_q, \quad (2-4)$$

where m_d and m_q are the modulation indices in the dq-frame. The modulation values are generated by the current controllers, which determine how the converter should switch in order to regulate the AC currents and, indirectly, the active and reactive power.

The power flowing through the rectifier can also be conveniently expressed in the dq-frame. The instantaneous input power from the AC side is:

$$p_{ac} = \frac{3}{2} (v_{s,d} i_d + v_{s,q} i_q). \quad (2-5)$$

Similarly, the power transferred by the converter into the DC-link is:

$$p_{conv} = \frac{3}{2} (v_{conv,d} i_d + v_{conv,q} i_q). \quad (2-6)$$

Assuming negligible losses between AC and DC sides, the DC-link current becomes:

$$i_{dc} = \frac{p_{conv}}{V_{dc}}. \quad (2-7)$$

This leads to the DC-link voltage dynamics:

$$C_{dc} \frac{dV_{dc}}{dt} = \frac{3}{2V_{dc}} (v_{conv,d} i_d + v_{conv,q} i_q) - i_{dc,out} - \frac{V_{dc}}{R_{dc}}. \quad (2-8)$$

This equation represents the energy balance in the rectifier: the capacitor voltage rises or falls depending on how much power is transferred from the AC source versus how much is drawn by the DC-side loads (inverters, battery chargers, etc.).

PWM rectifier typically employs a cascaded control architecture. The outer loop regulates the DC-link voltage by adjusting the reference for the d -axis current. Since active power is associated with i_d , any deviation in V_{dc} is corrected by increasing or

decreasing active power drawn from the AC source. The reactive power (or PCC voltage) is controlled by the q -axis current.

The inner current controllers force the actual currents to follow their references. Their control law is expressed as:

$$v_{conv,d}^* = K_{p,d}(i_d^* - i_d) + K_{i,d} \int (i_d^* - i_d) dt - \omega L_f i_q + v_{s,d}^{ff}, \quad (2-9)$$

$$v_{conv,q}^* = K_{p,q}(i_q^* - i_q) + K_{i,q} \int (i_q^* - i_q) dt + \omega L_f i_d + v_{s,q}^{ff}. \quad (2-10)$$

The feedforward terms $v_{s,d}^{ff}$ and $v_{s,q}^{ff}$ help compensate for the source voltage, thereby improving the transient response and reducing dependency on controller tuning. These equations form the complete dq-frame average model of a PWM rectifier and establish the analytical foundation for advanced control techniques such as VOC, DPC, power-factor correction, and DC-link stabilization within hybrid locomotive powertrains.

2.2.2 Self-Excited Synchronous Generator (SESG)

A self-excited synchronous generator (SESG) is a type of synchronous generator that utilizes its own generated voltage to excite the field winding, eliminating the need for an external power source for excitation. This characteristic makes SESGs particularly suitable for applications where external excitation sources are unavailable or impractical, such as in remote locations or standalone power systems. The main applications of SESGs in a hybrid powertrain system includes its primary characteristic of acting as the main electromechanical interface in diesel-electric and hybrid locomotive powertrains. In hybrid architecture, the SESGs not only supplies electrical power to the DC link but may also operate as a starter-generator when paired with a bidirectional converter. This dual-mode capability eliminates the need for mechanical cranking systems and significantly reduces battery cycling stress [13].

The SESGs provides regulated AC output through controlled field excitation and can quickly respond to sudden changes in the dynamic traction demands when supported by fast electrical excitation loops [5]. In real world applications, SESGs receives its mechanical input from the torque produced by the diesel engine, to crank the generator to produce electrical power, while its electrical output interacts with the rest of the system such as the PWM rectifier, DC link, inverter, Induction Motors, and the energy storage system (i.e batteries) [5].

The electromechanical dynamics of the SESG can be described by the following

set of differential equation:

$$\begin{aligned}
\frac{d\omega_r}{dt} &= \frac{1}{J}(T_m - T_e - B\omega_r) \\
\frac{d\delta}{dt} &= \omega_r - \omega_s \\
\frac{dE_f}{dt} &= \frac{1}{T_f}(V_f - E_f) \\
\frac{dI_d}{dt} &= \frac{1}{L_d}(V_d - R_s I_d + \omega_r L_q I_q) \\
\frac{dI_q}{dt} &= \frac{1}{L_q}(V_q - R_s I_q - \omega_r L_d I_d)
\end{aligned} \tag{2-11}$$

Where J is the rotor's inertia, T_m the mechanical torque input from the diesel engine prime mover, T_e is the electromagnetic torque produced by the generator, ω_r is the rotor angular velocity, and B is the damping coefficient that represents the mechanical losses [11].

2.2.2.1 3-Phase Source

The output generated by the SESG that suffice the bidirectional PWM converter is a three-phase AC voltage source. A three-phase voltage source is characterized by having an electrical installation of three phase wires and a neutral wire. The three-phase system is widely used in power generation and distribution around 380 to 480 V due to its efficiency in power transmission and ability to provide a constant power flow. In a three-phase system, the voltages in each phase are separated by 120 degrees, resulting in a balanced load and reduced vibration in motors [11].

A three-phase voltage source can be seen having a shift of 120 degrees between each phase. If V_{ph} is the phase voltage, then the equations for the three-phase voltages can be expressed as follows:

$$V_a(t) = V_{ph} \sin(\omega t) \tag{2-12}$$

$$V_b(t) = V_{ph} \sin(\omega t - \frac{2\pi}{3}) \tag{2-13}$$

$$V_c(t) = V_{ph} \sin(\omega t + \frac{2\pi}{3}) \tag{2-14}$$

2.2.2.2 Mathematical Model (Park Transformation)

Park transformation is a form of mathematical transformation that converts the electrical "ABC" reference frame (three-phase stationary frame) into a rotating reference frame that moves at the same angular speed as the system's base angular frequency or "dq0". This transformation simplifies the analysis of AC systems by producing constant values under steady-state conditions. In a balanced three-phase power system, the Park transformation converts the three-phase voltages (v_a, v_b, v_c) into direct-axis (v_d), quadrature-axis (v_q), and zero-sequence (v_0) components. The transformation is given by the following matrix equation:

$$T(\omega t_0) = \frac{2}{3} \begin{bmatrix} \cos(\omega t_0) & \cos\left(\omega t_0 - \frac{2\pi}{3}\right) & \cos\left(\omega t_0 + \frac{2\pi}{3}\right) \\ -\sin(\omega t_0) & -\sin\left(\omega t_0 - \frac{2\pi}{3}\right) & -\sin\left(\omega t_0 + \frac{2\pi}{3}\right) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad (2-14)$$

$$T(\omega t_0)^{-1} = \begin{bmatrix} \cos(\omega t_0) & -\sin(\omega t_0) & 1 \\ \cos\left(\omega t_0 - \frac{2\pi}{3}\right) & -\sin\left(\omega t_0 - \frac{2\pi}{3}\right) & 1 \\ \cos\left(\omega t_0 + \frac{2\pi}{3}\right) & -\sin\left(\omega t_0 + \frac{2\pi}{3}\right) & 1 \end{bmatrix} \quad (2-15)$$

$$f_{dq0} = T(\omega t_0)f_{abc}, \quad f_{abc} = T(\omega t_0)^{-1}f_{dq0}$$

the stator voltage equations in the dq0 reference frame can be expressed as follows:

$$v_d = R_s i_d + \frac{d\phi_d}{dt} - \omega_s \phi_q \quad (2-15)$$

$$v_q = R_s i_q + \frac{d\phi_q}{dt} + \omega_s \phi_d \quad (2-16)$$

with the flux linkages defined as:

$$\phi_d = L_d i_d + \phi_f \quad (2-17)$$

$$\phi_q = L_q i_q \quad (2-18)$$

Along the electromagnetic torque produced by the generator can be expressed as:

$$T_e = \frac{3}{2}P(\phi_d i_q - \phi_q i_d) \quad (2-19)$$

Where P is the number of poles. These equations form the basis for the gen-

erator to rectifier control design and for analyzing interactions between mechanical and electrical subsystems dynamics.

2.2.2.3 Voltage Regulation: Automatic Voltage Regulator (AVR) and Excitation machine

The Automatic Voltage Regulator (AVR) is a crucial component in the SESG system that maintains the stable terminal voltage V_t of the generator under varying load conditions by continuously adjusting the field excitation voltage V_f and the field rotor excitation current i_f . The AVR monitors the output voltage and compares it to a reference voltage V_{ref} [11]. Any deviation from the reference prompts the AVR to modify the excitation voltage accordingly. The AVR can be modeled using a first-order transfer function:

$$G_{AVR}(s) = \frac{K_{AVR}}{1 + sT_{AVR}} \quad (2-20)$$

Where K_A is the gain and T_{AVR} is the time constant of the AVR. The excitation system, which includes the exciter machine, converts the electrical power from the generator into the required DC voltage for field excitation.

2.2.2.4 Transformer and Rectifier Integration

In many hybrid locomotive interpretation, the output of the excitation system may be routed through a transformer for the voltage adaptation and isolation before entering the rectifier stage.

There are two types of transformer commonly use in electrical systems, as shown:

- Step-up: A transformer that recieves energy at low voltages and transforms it into a much higher output voltage [11].
- Step-down: A transformer that recieves a certain value of energy/voltage and decreasing the output allowing to be distributed into a many appliances etc. [11]

The rectifier may be an uncontrolled diode bridge, a thyristor-controlled, or a modern PWM active front end depending on system requirement, before advancing to the field voltage and current controller before finally becoming the input for the synchronous generator.

Figure 2.1. RL filter configuration between source and converter

2.2.2.5 Field Voltage and Current Connector

The field voltage and current connector provides the electrical interface between the AVR/excitation system and the rotor field winding. Its electrical dynamics are governed by:

$$v_f(t) = R_f i_f(t) + L_f \frac{di_f(t)}{dt} + e_{fm}(t) \quad (2-21)$$

Where R_f and L_f are the resistance and inductance of the exciter field winding, respectively, and $e_{fm}(t)$ is the back EMF induced in the exciter field winding. The AVR and exciter work in tandem to ensure that the generator maintains a stable output voltage, which is essential for the proper operation of the downstream power electronics and loads [4]. Modern Locomotives frequently employ static exciter or converter-fed brushless exciters, allowing fast response to transients such as sudden traction load changes or regenerative braking [11].

The complete excitation system can be characterized as:

$$i_f(s) = G_{exc}(s)[V_{ref}(s) - V_t(s)] \quad (2-22)$$

With G_{exc} capturing the exciter dynamics, amplifier behaviour, and limiting functions. The field connector must handle both generator-mode excitation and motor-mode support during cranking. Additionally, protection mechanisms—current limiting, thermal supervision, and open-field detection—are essential to prevent runaway terminal voltage or field under-excitation conditions. LC filtering may be applied to suppress high-frequency switching noise originating from converter-fed excitation systems.

2.2.3 RL Filter

RL filter serves as a fundamental passive filtering solution for mitigating high-frequency noise and harmonic content in power electronic systems. Comprising a series resistor (R) and inductor (L), this filter topology provides effective attenuation of switching harmonics while offering simplicity and reliability in implementation. RL filters are particularly advantageous in applications requiring current smoothing and limited high-frequency noise suppression without the complexity of more elaborate filter structures [14].

The primary function of an RL filter lies in its ability to provide a first-order low-pass characteristic, effectively damping high-frequency components while allowing

fundamental frequency signals to pass with minimal attenuation. The design methodology for RL filters involves careful consideration of both filtering performance and system efficiency, as the resistive element introduces additional losses.

Designing an RL filter involves the following systematic procedure:

- **Inductor selection (L):** The inductor value is determined based on the required attenuation at the switching frequency and the maximum allowable current ripple. The fundamental relationship governing inductor selection is given by:

$$L = \frac{R_L}{\sqrt{\left(\frac{f_{sw}}{f_c}\right)^2 - 1}} \quad (2-23)$$

where R_L represents the load resistance, f_{sw} is the switching frequency, and f_c denotes the desired cutoff frequency. Alternatively, for current ripple consideration:

$$L = \frac{V_{max}}{\Delta I_{max} \cdot f_{sw}} \quad (2-24)$$

where V_{max} is the maximum voltage across the inductor and ΔI_{max} is the maximum allowable current ripple [12].

- **Resistor selection (R):** The resistor value is optimized to balance between damping effectiveness and power loss. The optimal damping resistance can be determined by:

$$R = 2\zeta \sqrt{\frac{L}{C_{eq}}} \quad (2-25)$$

where ζ represents the damping ratio (typically between 0.7 and 1.0 for critical damping) and C_{eq} is the equivalent capacitance of the system. The power dissipation in the resistor must be carefully evaluated:

$$P_R = I_{rms}^2 \cdot R \quad (2-26)$$

where I_{rms} is the RMS current through the filter [15].

- **Frequency response verification:** The complete filter transfer function must be verified to ensure proper attenuation characteristics:

$$H(s) = \frac{1}{1 + \frac{R}{L}s} \quad (2-27)$$

The cutoff frequency should satisfy $f_{grid} \ll f_c \ll f_{sw}$ to ensure effective harmonic suppression while maintaining fundamental frequency performance.

2.2.4 LCL Filter

As an effort to further diminishing the harmonics of the voltage, current, and any other signal carried from the source generator to the rectifier, an LCL filter is used. LCL filter itself is a passive filtering component that comprises of two inductors, one source inductor L_1 , a parallel capacitor C , lastly the converter side inductor L_2 . These types of filtering are commonly use in an active rectifier system or a general grid system. LCL filters are able to dampen the harmonics of the switching frequency at a greater effieciency than both L and LC filter, along with a more compact form of circuitry [14].

The main function of an LCL filter lies is its ability to reduce the high frequency ripple of the current and suppresses the Total Harmonic Distortion (THD) of the source current. Despite the advantages that the LCL brought, it also brought the drawbacks of creating a resonating frequency that can go out of hand if left uncontrolled and unchecked. Designing an LCL filter involves the measurement and determining the value of the variable, using these procedure below:

- **Converter-side inductor (L_1):** The value of L_1 is decided based on the maximum current ripple that is allowed. Using the equation below, the value of L_1 can be written as:

$$L_1 = \frac{V_{DC}}{6\Delta I_{L,max}f_{sw}} \quad (2-28)$$

With V_{DC} being the voltage of the DC-Link, $\Delta I_{L,max}$ is the allowed maximum current ripple that are going to be used, and f_{sw} being the switching frequency of the converter.

- **Filtering capacitor (C):** The value of C is proportionally limited to the reactive power the grid recieves, with the average being no more than 5% of the nominal power of the system. The value of C can be determined using these equation:

$$C = 0.05C_b \quad (2-29)$$

$$C_b = \frac{1}{\omega_n Z_b} \quad (2-30)$$

$$Z_b = \frac{E_n^2}{P_n} \quad (2-31)$$

With the C_b being the capacitance of the base system, ω_n as the nominal angular frequency of the source ($\omega_n = 2\pi f_{source}$). Z_b is the base impedance

Figure 2.2. RC filter configuration for DC-link voltage smoothing

of the system along with E_n being the nominal voltage, and P_n as the nominal power.

- **Source-side inductor:** Lastly, the value of source focused inductor is determined by the ratio of the desired harmonics attenuation from the converter connected inductor. Using the equation below, L_2 can be determine as:

$$L_2 = rL_1, 0.5 \leq r \leq 1 \quad (2-32)$$

The value presented needs to be further verified as the resonance frequency fulfills the $10f_{grid} < f_{res} < 0.5f_{sw}$ and the total inductance does not exceed 0.1 p.u. to prevent any voltage drops.

2.2.5 RC filtering for DC-Link

The RC filter represents a fundamental passive filtering approach specifically designed for voltage smoothing in DC-link applications. Consisting of a series resistor (R) and parallel capacitor (C), this configuration provides effective attenuation of voltage ripples and high-frequency noise generated by switching converters. RC filters are extensively employed in DC-link circuits for their simplicity, cost-effectiveness, and reliable performance in suppressing voltage harmonics [14].

The principal function of an RC filter in DC-link applications is to maintain stable DC voltage by filtering out switching frequency components and preventing their propagation to sensitive load circuits. The capacitor stores energy during voltage peaks and releases it during troughs, while the resistor provides damping to prevent oscillations and limit inrush currents.

The design methodology for RC filters in DC-link applications follows this systematic approach:

- **Capacitor selection (C):** The capacitance value is primarily determined by the required voltage ripple specification and the system dynamics. The fundamental design equation is given by:

$$C = \frac{\Delta I_{max}}{8 \cdot f_{sw} \cdot \Delta V_{max}} \quad (2-33)$$

where ΔI_{max} represents the maximum current ripple, f_{sw} is the switching frequency, and ΔV_{max} denotes the maximum allowable voltage ripple. For

energy storage considerations:

$$C = \frac{2 \cdot P \cdot \Delta t}{\Delta V^2} \quad (2-34)$$

where P is the power level, Δt is the hold-up time requirement, and ΔV is the acceptable voltage drop [16].

- **Resistor selection (R):** The series resistance is optimized to provide adequate damping while minimizing power losses. The optimal resistance can be determined by:

$$R = \frac{1}{2\pi \cdot f_c \cdot C} \quad (2-35)$$

where f_c is the desired cutoff frequency. The power dissipation constraint must be considered:

$$R \leq \frac{\Delta V_{max}}{I_{max}} \quad (2-36)$$

where I_{max} is the maximum current through the filter.

- **Time constant verification:** The RC time constant must be verified to ensure proper filtering characteristics:

$$\tau = R \cdot C = \frac{1}{2\pi \cdot f_c} \quad (2-37)$$

The cutoff frequency should be selected such that $f_c \ll f_{sw}$ to effectively attenuate switching harmonics while maintaining DC voltage stability. The transfer function characterization:

$$H(s) = \frac{1}{1 + RCs} \quad (2-38)$$

provides the theoretical foundation for frequency response analysis [13].

2.2.6 Phase Locked Loop (PLL)

The main objective of the Phase-Locked Loop (PLL) is to track the angle of the input voltage, this angle is then used for synchronization. Phasor or three-phase representation of the voltage is converted / passed to dq reference frame. For dq modeling, θ is the angle between converter's frame and the source's frame [14]. q -axis signal is enforced to zero through PI control, which then results the frequency deviation. This frequency will be integrated to obtain voltage angle

2.2.6.1 Voltage and Current oriented Control (VoC)

2.2.7 Traction Battery

Traction batteries are high-energy and high-power rechargeable battery systems designed to provide propulsion energy for electric vehicles (EVs and HEVs). A traction battery pack consists of many cells arranged in series and parallel to achieve the required voltage, power, and energy capacity. Because cells exhibit variations in capacity, internal resistance, open-circuit voltage (OCV), and aging rates, the overall performance of the pack is often limited by the weakest cell. Therefore, battery management strategies including monitoring, modelling, and balancing are necessary to ensure safety, maximize usable energy, and prolong battery life [?].

Key characteristics relevant to traction applications include energy density, acceptable C-rates for charging and discharging, thermal limits, and pack architecture (series/parallel configuration and sensing topology). These aspects directly influence BMS design and dictate the necessity for reliable SoC estimation and cell balancing [?].

2.2.7.1 Model Equivalent Circuit

Equivalent-circuit models (ECMs) are widely used in traction-battery BMS because they offer a practical balance between accuracy and computational efficiency. Several ECM families exist:

1. R_s + OCV Model. The simplest model represents the terminal voltage as

$$V_t = \text{OCV}(\text{SOC}) - R_s I,$$

where R_s is the instantaneous series resistance. This model is suited only for coarse estimations due to its inability to represent transient dynamics [?].

2. Thevenin (R_s + RC) Model. A more accurate ECM is the first-order Thevenin model, which augments the series resistance with a parallel R_p C_p network capturing polarization and short-term transients:

$$V_t(s) = \text{OCV}(\text{SOC}) - I(s) \left(R_s + \frac{R_p}{1 + sR_pC_p} \right).$$

In time domain, this leads to the widely used recursive voltage-update form implemented in embedded BMS systems. This model accurately captures dynamic behavior while

remaining computationally efficient, making it a preferred choice for Kalman-filter-based SoC estimation [?].

3. Multi-RC / Impedance-based Models. More sophisticated models incorporate multiple RC branches or Warburg (diffusion) elements to better capture multi-scale dynamic processes. These models offer higher fidelity but at the cost of increased parameterization effort and computation [?].

Parameter Identification. Typical identification procedures include:

- standard CC CV cycling for capacity measurement,
- stepped discharge tests with relaxation periods to obtain the OCV SoC curve,
- DCIR tests or electrochemical impedance spectroscopy (EIS) for resistance and polarization-parameter extraction.

Relaxation curves are commonly used to estimate the RC time constants for Thevenin models [?].

2.2.7.2 SoC, OCV, Internal Resistance

State of Charge (SoC). State of Charge represents the remaining charge in a cell normalized by its rated capacity:

$$SOC(t) = SOC(t_0) - \frac{1}{Q_{\text{rated}}} \int_{t_0}^t I(\tau) d\tau,$$

with the sign convention depending on whether current is defined positive during charge or discharge. Coulomb counting is simple and offers high-resolution short-term tracking but suffers from drift caused by sensor errors and capacity aging, necessitating periodic correction via voltage-based methods [?].

Open-Circuit Voltage (OCV). The OCV is the equilibrium voltage of a cell at zero current. Lithium-ion chemistries exhibit nonlinear OCV SoC characteristics typically modeled through lookup tables, polynomials, or spline fits derived from incremental discharge and relaxation experiments. Because OCV provides a direct observable for SoC, it is used to correct Coulomb-counter drift. However, long rest times are generally required for accurate OCV measurement, motivating dynamic voltage-correction methods within state estimators [?, ?].

Internal Resistance (IR). Internal resistance includes the instantaneous series resistance R_s and the polarization-related resistances R_p from the ECM. IR influences the voltage drop under load and contributes to heat generation. Since IR varies with SoC, temperature, and aging, accurate modeling is essential for predicting power capability and improving SoC estimation via compensation of load-induced voltage drops [?, ?].

SoC Estimation via ECM and Filtering. Because terminal voltage is influenced by OCV, internal resistance, and dynamic polarization, model-based estimation approaches are typically employed. Extended Kalman filters (EKF), unscented Kalman filters (UKF), adaptive EKF (AEKF), and particle filters combine the ECM with measurements to estimate SoC and, in some cases, parameters such as R_s , R_p , or effective capacity. A common state vector is

$$x = [SOC, V_{RC1}, V_{RC2}, \theta]^T,$$

where the V_{RC} states represent RC-branch voltages and θ denotes time-varying parameters. Such methods improve robustness against noise, aging, and operating-condition variability [?, ?].

2.2.7.3 Cell Balancing: Passive and Active Methods

Cells connected in series develop SoC and voltage imbalance due to differences in capacity, internal resistance, temperature, and degradation rates. Balancing is required to prevent overcharge or deep discharge of weak cells and to maximize usable pack capacity.

Passive Balancing. Passive (dissipative) balancing removes excess charge from higher-energy cells through resistive paths that dissipate energy as heat. This method is simple and low-cost but slow and energy-inefficient, making it less suitable for large traction packs [?].

Active Balancing. Active balancing redistributes energy among cells or between cells and an intermediate storage medium (capacitor, inductor, or redundant cell). Major topologies include cell-to-cell, cell-to-pack, pack-to-cell, and hybrid structures. Active systems provide faster equalization and higher efficiency but at the expense of increased circuit complexity and cost [?].

2.2.8 VVVF Inverter

A Variable Voltage Variable Frequency (VVVF) inverter commonly implemented as a three-phase Voltage Source Converter (VSC) employing PWM switching is the primary power electronic interface between the DC link (or generator/rectifier stage) and the traction motors in modern powertrains. The VVVF inverter's purpose is to synthesize controlled three-phase AC voltages whose amplitude and frequency are varied according to the desired motor speed and torque profile. Compared to simple open-loop V/f schemes, modern closed-loop VVVF implementations (using SPWM, SVPWM or other modulation methods) provide superior voltage utilization, precise torque control, regenerative capability, and improved dynamic performance required by traction applications [17]. Vector-control implementations (FOC/IFOC) operating on the VVVF inverter allow the induction motor to be controlled as if it were a separately-excited DC machine, greatly simplifying speed controller design and enabling high dynamic torque response [18]

To meet the dual objectives of high dynamic performance and robustness in traction environments, VVVF inverters are generally designed with the following attributes:

- **High voltage utilization and modulation flexibility:** Modulation schemes such as SPWM (with third-harmonic injection), SVPWM, and over-modulation PWM are used to maximize fundamental voltage magnitude and minimize harmonic content while staying within the DC-link limits. Over-modulation and 60° PWM can increase fundamental voltage beyond the linear SPWM limit when needed (e.g., high torque or high speed), at the cost of increased harmonic content which must be managed. Practical traction drives selectively switch modulation regions depending on required voltage utilization and acceptable distortion. [17].
- **Bidirectional energy flow (regeneration):** The inverter is fully bidirectional, allowing motoring (DC to AC) and generating (AC to DC) modes. Properly controlled, the inverter returns braking energy to the DC bus or upstream storage and supports regenerative braking strategies in hybrid and electric trains [17].
- **Fast torque control via vector control:** When combined with field-oriented control (flux and speed feedback, flux observers), the inverter can enforce decoupled torque and flux commands (i_q and i_d), enabling rapid torque transients and accurate speed regulation under variable loads [18]
- **Robustness to stability and oscillations:** Cascaded VVVF inverter motor systems can exhibit low-frequency oscillations (particularly under light/no-load and certain control parameterizations). Small-signal impedance-based

stability analysis (Middlebrook-style impedance ratio and state-space linearization) is therefore commonly used to detect and mitigate oscillatory bands (e.g., 15-45 Hz in typical examples) and to design damping or controller changes to ensure stable operation in multi-machine and DC/AC cascaded topologies [8].

In traction converters the VVVF inverter is typically implemented as a three-phase bridge formed by six IGBT (or SiC MOSFET) switches with appropriate dead-time handling, current sensing, and a DC-link capacitor. An optional AC side filter (L or LCL) is frequently installed between the inverter and the motor (or traction transformer) to limit switching ripple and to shape the converter input impedance for stability reasons [8].

2.2.8.1 Operating Modes and Modulation

Common modulation/mode choices and their practical role in VVVF drives are:

- **SPWM (Sinusoidal PWM)** — simplest to implement, linear modulation up to modulation index $M_a \leq 1$, good waveform quality in linear region. Used where cost and simplicity are prioritized. [17]
- **SVPWM (Space-Vector PWM)** — better DC-link utilization ($\sim 15.5\%$ higher fundamental than linear SPWM) and lower switching distortion for a given switching frequency; popular in high-performance traction drives.
- **Over-modulation / 60° PWM** — employed when the converter must extract the maximum possible fundamental voltage from the DC bus (e.g., near maximum speed). 60° PWM flattop portions of the waveform to raise fundamental magnitude ($\approx 1.15 \times$ SPWM) but increases harmonics; often used in controlled manner for short periods. [17]
- **Linear vs. non-linear regions** — control stacks switch between linear PWM and over-modulation regions according to commanded frequency and voltage demand to avoid saturation while keeping overall distortion within acceptable bounds.

2.2.8.2 Control Architecture (Functional Overview)

A typical high-performance VVVF inverter control stack for traction contains:

- **Outer Speed/Flux Loops:** Speed PI regulator (outer loop) provides torque (or i_q^*) command; flux control (direct or via flux observer) provides i_d^* reference. In vector control, the induction motor behaves like a separately excited DC motor; hence PI tuning methods from DC control are commonly used to design the speed regulator. [18]

- **Current (Inner) Loops:** Fast current controllers (usually PI) regulate i_d and i_q to their references, including cross-coupling compensation and feedforward terms to cancel ωL coupling effects. The current loop runs at a much faster bandwidth than the speed loop to ensure fast torque dynamics.
- **Modulator / PWM:** The current controller outputs modulation references (m_a, m_b, m_c) or equivalent dq modulation indices (m_d, m_q) which the PWM module converts to gate pulses for the semiconductor switches (taking into account dead-time, over-modulation transition, and carrier frequency).
- **Flux Observer / Estimator:** Especially for sensorless operation or robust flux feedback, a flux observer (model-based or sliding-mode type) provides rotor/stator flux estimation to ensure accurate field orientation and decoupling. Flux feedback improves dynamic performance and reduces sensitivity to parameter mismatch. [18]
- **Protection and Diagnostics:** Overcurrent, overvoltage, thermal monitoring, and torque/adhesion limiters are integrated to ensure safe traction operation.

2.2.8.3 Mathematical Model of the VVVF Inverter in the dq -Frame

For analysis and controller design the VVVF inverter and its interface to the motor (or filter) are commonly expressed in the rotating dq frame (Park transform), aligned to the motor flux or stator voltage vector depending on the chosen control strategy. The average (fundamental) model of the inverter neglects switching ripple and replaces the switching action by equivalent averaged converter voltages. The following model assumes an $L_f - R_f$ filter (or stator leakage equivalent) between converter terminals and the motor:

$$\begin{aligned} v_{conv,d} &= v_{x,d} + R_f i_d + L_f \frac{di_d}{dt} - \omega L_f i_q, \\ v_{conv,q} &= v_{x,q} + R_f i_q + L_f \frac{di_q}{dt} + \omega L_f i_d, \end{aligned} \quad (2-39)$$

where $v_{conv,d}, v_{conv,q}$ are the converter (averaged) voltages in dq frame, $v_{x,d}, v_{x,q}$ are the motor terminal (or load) voltages in dq frame, i_d, i_q are filter/motor currents, and ω is the angular speed of the chosen reference frame. The cross-coupling terms $\pm \omega L_f i_{q,d}$ arise due to rotating coordinates and must be compensated in high-performance controllers.

The converter averaged voltage is commonly written in terms of DC-link voltage V_{dc} and modulation indices m_d, m_q (dq projection of the space vector):

$$v_{conv,d} = \frac{V_{dc}}{2} m_d, \quad v_{conv,q} = \frac{V_{dc}}{2} m_q. \quad (2-40)$$

(For two-level VSC the $V_{dc}/2$ factor maps a normalized modulation vector to phase voltages; the exact constant depends on modulation convention and transform normalization.)

The instantaneous electromagnetic power delivered to the motor terminals from the converter in dq coordinates is:

$$p_{conv} = \frac{3}{2}(v_{conv,d}i_d + v_{conv,q}i_q). \quad (2-41)$$

and the DC-link capacitor energy balance (assuming DC-link capacitance C_{dc} and DC current out $i_{dc,out}$) is:

$$C_{dc} \frac{dV_{dc}}{dt} = \frac{3}{2V_{dc}}(v_{conv,d}i_d + v_{conv,q}i_q) - i_{dc,out} - \frac{V_{dc}}{R_{dc}}. \quad (2-42)$$

2.2.8.4 Controller Structure and Decoupling

To achieve independent active/reactive (or torque/flux) regulation, the current controller synthesizes converter voltage commands using PI control with decoupling feedforward terms. A typical control law (average model) in dq frame is:

$$v_{conv,d}^* = K_{p,d}(i_d^* - i_d) + K_{i,d} \int (i_d^* - i_d) dt - \omega L_f i_q + v_{x,d}^{ff}. \quad (2-43)$$

$$v_{conv,q}^* = K_{p,q}(i_q^* - i_q) + K_{i,q} \int (i_q^* - i_q) dt + \omega L_f i_d + v_{x,q}^{ff}. \quad (2-44)$$

The $\pm\omega L_f$ terms cancel the rotating frame cross coupling, while $v_{x,d}^{ff}, v_{x,q}^{ff}$ are feedforward terms (e.g., measured terminal voltages or estimated motor back-EMF) that improve transient response. These controller outputs are then converted into modulation indices (m_d, m_q) and finally to PWM gate signals, accounting for modulation limits and over-modulation transitions [18] [17].

2.2.8.5 Average Converter Impedance and Small-Signal Stability

For multi-machine or multi-converter installations (and to study tendency to low-frequency oscillations), it is useful to derive the small-signal input impedance of the inverter (in dq0 frame) and compare it with the motor (or load) impedance using Middlebrook-type impedance ratio criteria. The inverter's small-signal input/output impedance matrices $Z_{s,dq}(s)$ and the motor/load admittance $Y_{m,dq}(s)$ form the cascade stability loop:

$$\text{Loop gain (dq)} = Z_{s,dq}(s)Y_{m,dq}(s) \quad (2-45)$$

If the generalized Nyquist (or matrix impedance) criterion indicates encirclement

of -1 in the relevant frequency band, low-frequency oscillations may occur. Studies show that for particular controller gains and filter configurations, the cascade can be prone to oscillation in low-frequency bands (e.g., 15–45 Hz), so impedance shaping (via filter design, virtual damping, or controller retuning) is often required in traction VVVF designs [8].

2.2.9 Asynchronous Machine (Induction Motor)

Sitting at the edge of the system, acting as the final bridge and component from mechanical to electrical and back to mechanical. The induction motor (IM), or asynchronous machine, is an AC electric motor where torque is produced by the interaction of a rotating magnetic field and currents induced in the rotor by electromagnetic induction. Its robustness, low cost, and minimal maintenance needs make it a cornerstone of industrial drives.

2.2.9.1 Operating Principle of the Induction Motor

The operation of a three-phase induction motor is based on the following fundamental principles:

1. **Generation of a Rotating Magnetic Field:** When a balanced three-phase AC supply is applied to the spatially displaced stator windings, it generates a magnetic field that rotates at a constant *synchronous speed* (ω_e). This speed is given by:

$$\omega_e = \frac{4\pi f}{P}$$

where f is the supply frequency and P is the number of poles.

2. **Induction of EMF in the Rotor:** The rotating stator magnetic field cuts the stationary conductors of the rotor, inducing an electromotive force (EMF) according to Faraday's law. In a squirrel-cage motor, these conductors are short-circuited by end rings, causing currents to flow.
3. **Generation of Torque:** The current-carrying rotor conductors are situated within the stator's magnetic field. A mechanical force (the Lorentz force) acts on them, producing a torque that accelerates the rotor in the direction of the rotating field. The rotor speed (ω_r) always remains less than the synchronous speed.

The difference between the synchronous and rotor speeds is quantified by the **slip** (s):

$$s = \frac{\omega_e - \omega_r}{\omega_e}$$

At standstill ($\omega_r = 0$), the slip is 1; at synchronous speed (no load), the slip approaches 0. The motor operates in the region between these two points.

2.2.9.2 dq Model of the Induction Motor

The dynamic model in the natural **abc**-reference frame involves time-varying mutual inductances due to rotor movement, leading to complex differential equations. The **dq-reference frame transformation** (Park's Transformation) simplifies this by converting the three-phase system into an orthogonal, two-axis system rotating at an arbitrary speed ω , where the inductances become constant.

The transformation from **abc** variables to **dq** variables is given by:

$$\mathbf{f}_{dqns}^\omega = \mathbf{T}(\theta)\mathbf{f}_{abcs}$$

where the transformation matrix $\mathbf{T}(\theta)$ is:

$$\mathbf{T}(\theta) = \frac{2}{3} \begin{bmatrix} \cos \theta & \cos(\theta - \frac{2\pi}{3}) & \cos(\theta + \frac{2\pi}{3}) \\ -\sin \theta & -\sin(\theta - \frac{2\pi}{3}) & -\sin(\theta + \frac{2\pi}{3}) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad \text{with} \quad \theta = \int \omega(\tau) d\tau + \theta(0)$$

The resulting **dq-model** for a squirrel-cage induction motor is as follows:

1. Voltage Equations (in a frame rotating at speed ω):

$$\begin{aligned} v_{ds}^\omega &= R_s i_{ds}^\omega + \frac{d\lambda_{ds}^\omega}{dt} - \omega \lambda_{qs}^\omega \\ v_{qs}^\omega &= R_s i_{qs}^\omega + \frac{d\lambda_{qs}^\omega}{dt} + \omega \lambda_{ds}^\omega \\ 0 &= R_r i_{dr}^\omega + \frac{d\lambda_{dr}^\omega}{dt} - (\omega - \omega_r) \lambda_{qr}^\omega \\ 0 &= R_r i_{qr}^\omega + \frac{d\lambda_{qr}^\omega}{dt} + (\omega - \omega_r) \lambda_{dr}^\omega \end{aligned}$$

The terms $\omega \lambda_{qs}$ and $\omega \lambda_{ds}$ are *speed voltages* arising from the rotation of the reference frame.

2. Flux Linkage Equations (Become Constant and Decoupled):

$$\begin{aligned} \lambda_{ds}^\omega &= L_s i_{ds}^\omega + L_m i_{dr}^\omega & \lambda_{qs}^\omega &= L_s i_{qs}^\omega + L_m i_{qr}^\omega \\ \lambda_{dr}^\omega &= L_r i_{dr}^\omega + L_m i_{ds}^\omega & \lambda_{qr}^\omega &= L_r i_{qr}^\omega + L_m i_{qs}^\omega \end{aligned}$$

where $L_s = L_{ls} + L_m$ and $L_r = L_{lr} + L_m$. L_m is the magnetizing inductance.

3. Torque Equation: The electromagnetic torque can be expressed in several forms,

a common one being:

$$T_e = \frac{3}{2} \frac{P}{2} L_m (i_{qs}^\omega i_{dr}^\omega - i_{ds}^\omega i_{qr}^\omega)$$

Alternatively, in terms of rotor flux:

$$T_e = \frac{3}{2} \frac{P}{2} \frac{L_m}{L_r} (\lambda_{dr}^\omega i_{qs}^\omega - \lambda_{qr}^\omega i_{ds}^\omega)$$

The choice of the reference frame speed ω yields different models:

- **Stationary Reference Frame** ($\omega = 0$): The dq-axes are fixed to the stator.
- **Synchronous Reference Frame** ($\omega = \omega_e$): The dq-axes rotate with the stator field. Steady-state variables become DC quantities, which is ideal for control.
- **Rotor Reference Frame** ($\omega = \omega_r$): The dq-axes are fixed to the rotor.

2.2.9.3 Field-Oriented Control (FOC)

Field-Oriented Control (FOC), or Vector Control, is an advanced technique that enables instantaneous torque control of an induction motor by decoupling the control of flux and torque, mimicking the performance of a separately excited DC motor.

1. **Concept** In a separately excited DC motor, torque is given by:

$$T = k \phi_f i_a$$

where the field flux ϕ_f (controlled by i_f) and the armature current i_a are independent and their magnetic fields are naturally maintained at 90° by the commutator. FOC creates an analogous condition in the induction motor by:

- Assigning the stator current component i_{ds}^e as the **flux-producing current** (analogous to i_f).
 - Assigning the stator current component i_{qs}^e as the **torque-producing current** (analogous to i_a).
 - Ensuring these two current vectors are always maintained 90° apart electrically.
2. **Rotor Flux Orientation (RFO)** To achieve decoupling, the **d**-axis of the synchronous reference frame (d^e - q^e) is aligned with the rotor flux vector λ_r . This condition is known as *field orientation*:

$$\lambda_{qr}^e = 0 \quad \text{and} \quad \lambda_{dr}^e = |\vec{\lambda}_r|$$

Under this condition, the motor model simplifies dramatically:

- **Flux-Current Relationship:**

$$\lambda_{dr}^e = \frac{L_m}{1 + T_r p} i_{ds}^e \quad \text{where} \quad T_r = \frac{L_r}{R_r} \quad (\text{Rotor Time Constant})$$

In steady-state, $\lambda_{dr}^e = L_m i_{ds}^e$. The rotor flux is directly controlled by i_{ds}^e .

- **Torque Equation:**

$$T_e = \frac{3P}{2} \frac{L_m}{L_r} \lambda_{dr}^e i_{qs}^e$$

By keeping λ_{dr}^e constant, the torque becomes directly and instantaneously proportional to i_{qs}^e :

$$T_e = K_T i_{qs}^e \quad \text{where} \quad K_T = \frac{3P}{2} \frac{L_m}{L_r} \lambda_{dr}^e$$

This is identical in form to the DC motor torque equation.

3. **Implementation: Direct and Indirect FOC** The implementation of FOC is categorized based on how the critical **rotor flux angle** θ_e is obtained.

- **Direct FOC:** The rotor flux is directly estimated or measured to calculate its angle.

$$\theta_e = \tan^{-1} \left(\frac{\lambda_{qr}^s}{\lambda_{dr}^s} \right)$$

The stationary frame stator flux λ_{dqs}^s is estimated by integrating the stator EMF ($\lambda = \int (v - Ri) dt$), from which the rotor flux λ_{dqr}^s is calculated. This method is robust to rotor parameter variations but performs poorly at low speeds due to integration drift and sensitivity to stator resistance.

- **Indirect FOC:** The flux angle is calculated indirectly using the slip relation required for field orientation.

$$\omega_{sl} = \frac{1}{T_r} \frac{i_{qs}^{e*}}{i_{ds}^{e*}} = \frac{R_r}{L_r} \frac{i_{qs}^{e*}}{i_{ds}^{e*}}$$

This is the fundamental equation of indirect FOC. The flux angle is then obtained by integrating:

$$\theta_e = \int \omega_e dt = \int (\omega_{sl} + \omega_r) dt$$

This method is easier to implement and works well at low speeds. However, its performance is **highly sensitive to the rotor time con-**

stant T_r . Inaccurate knowledge of T_r causes *detuning*, where the d-q axes are misaligned, leading to incorrect flux and torque production and a loss of decoupling.

4. **FOC System Control Structure** A standard Indirect FOC drive system consists of the following nested control loops:

- (a) **Speed Controller (PI):** Compares the speed reference ω_m^* with the feedback ω_m and generates a torque reference T_e^* .
- (b) **Flux Controller (PI):** Maintains the rotor flux magnitude λ_{dr}^e at its rated value. The output of this controller is the flux-producing current reference i_{ds}^{e*} .
- (c) **Current Command Calculation:** The torque-producing current reference i_{qs}^{e*} is calculated from T_e^* using the FOC torque equation. The flux-producing current reference i_{ds}^{e*} is either a constant or the output of the flux controller.
- (d) **Flux Angle Generation:** The flux angle θ_e is generated by integrating the sum of ω_{sl}^* and the measured rotor speed ω_r .
- (e) **Inverse Park Transform:** Transforms the synchronous current references i_{ds}^{e*} and i_{qs}^{e*} into stationary two-phase current references $i_{\alpha\beta}^{s*}$.
- (f) **2-Phase to 3-Phase Transform (Inverse Clarke):** Transforms $i_{\alpha\beta}^{s*}$ into three-phase current references i_{abc}^* .
- (g) **Current Regulator & PWM Inverter:** A current controller (typically a PI regulator in the synchronous frame) forces the motor's stator currents to instantaneously follow the references i_{abc}^* via a PWM-controlled voltage source inverter.

Through this structure, high-performance, instantaneous torque control of an induction motor, equivalent to that of a DC motor, is achieved.

CHAPTER III

RESEARCH METHODOLOGY

3.1 Final Project Tools

Tools and software that were used in the creation of this thesis consist of the following

1. **ASUS ROG Strix G814JV DESKTOP-RQVL4P8 Laptop** with the specifications: Windows 11 OS, 13th Gen Intel(R) Core(TM) i7-13650HX (2.60 GHz) processor, 16.0 GB of installed RAM, Hybrid storage 2 TB SSD and 451 GB HDD.
2. **MATLAB 2025a**, used to build and simulate the model using programming script and block diagram in Simulink.
3. **Lucidchart**, used to draw the electrical circuits, block-diagrams, and flowchart that are used in this final project.

3.2 Final Project Research Method: Modelling and Simulation

3.2.1 Synchronous Generator Data Parameters and Specifications

The synchronous generator modeled in this final project is a self excited brushless synchronous generator, which are moderately used in a modelling of a three-phase powertrain system. The specification and parameters that were used as models are detailed below:

Parameter	Value
Power Rating	2 MVA
Nominal Frequency	50 Hz
Terminal Voltage	680 V AC
Nominal Voltage	400 V AC
Nominal Field Current	100 A
Maximum Angular Velocity	1500 rpm

Table 3.1. Generator Specifications

From the parameter and specifications stated, the usage of nominal power and terminal voltage are used to calculate the first batch of basis variables that are going to be used for the remainder of this thesis:

$$S_{base} = 2MVA, V_{base} = 680V \quad (3-1)$$

$$Z_{base} = \frac{V_{base}^2}{S_{base}} \quad (3-2)$$

$$I_{base} = \frac{S_{base}}{\sqrt{3}V_{base}} \quad (3-3)$$

$$\omega_{base} = 2\pi f_n \quad (3-4)$$

$$L_{base} = \frac{Z_{base}}{\omega_{base}} \quad (3-5)$$

$$C_{base} = \frac{1}{Z_{base}\omega_{base}} \quad (3-6)$$

using the equations above from (3-1) and (3-2), the value of each base parameter is:

$$\begin{aligned} Z_{base} &= 0.23186\Omega \\ I_{base} &= 1698A \\ \omega_{base} &= 314.159rad/s \\ L_{base} &= 7.32 \times 10^{-3}H \\ C_{base} &= 1.37285 \times 10^{-2}F \end{aligned} \quad (3-7)$$

3.2.2 Rectifier Data Parameters

During the arrangement of this thesis, the rectifier that was used was a bidirectional AC to DC mosfet/IGBT rectifier, in which were commonly found and used in the modelling and simulation of three-phase system. Below are the following specification and parameter data that were used:

Parameter	Value
Power Rating	2 MVA
Nominal Frequency	50 Hz
Input Voltage	680 V AC
Output Voltage	1500 V DC
Maximum Rated Current	1698 A
Switching Frequency	10 kHz

Table 3.2. Rectifier Specifications

From the parameter and spesifications stated, the swiching frequency is used to compute the ω_{sw} that acts as the basis for both rectifier and inverter in this thesis:

$$\omega_{sw} = 2\pi f_{sw} \quad (3-8)$$

$$\omega_{sw} = 62,800 \text{ rad/s} \quad (3-9)$$

3.2.2.1 Rectifier Control Model

The control of the rectifier is modeled in MATLAB and Simulink using the approach in terms of differential equation and based on the assumption that the input provided by the generator is steady throughout the entire simulation. The m.file define and provides the parameters and variable that are used for the system, while the slx.file handle the arrangement of the control blocks and the computations of the system's variables.

3.2.2.2 PWM Converter Control

There are two primary control methods used in this final project, namely Voltage Oriented Control and Field Oriented Control. These methods are calculated based on each fundamental concept of their respective differential equations, and their connection to power electronics.

Voltage Oriented Control is a synchrononous-frame control strategy used for three-phase PMW and other grid-tied converters. VoC's key principle is to align the rotating dq reference frame with the generators voltage vector such that the d-axis corresponds to the active-power (real power) channel and the q-axis corresponds to the reactive-power channel. The VoC method is realized through three main procedure:

- **Sensing and Transformation** of the generator's three phase voltages and currents from the standard stationary ABC frame into the rotational dq frame.
- The use of a **Cascaded Control Structure**, where the DC-link voltage loop generates the d-axis voltage reference.
- **Decoupled Current Regulation**, where the dq current loop produces the dq voltage commands needed for the PWM modulator

3.2.2.3 Voltage Control Loop (Outer Loop)

The voltage control loop is implemented as the outer layer of the cascaded control structure. V_{dc} is measured and compared to the reference voltage (V_{ref}), resulting in the creation of (V_{error}) which will be processed by a discrete PI controller. Once controlled, the outputted value is directly used as the d-axis current reference i_d^* .

$$i_d^* = K_{pv}(V_{ref} - V_{dc}) + K_{iv} \int ((V_{ref} - V_{dc}))dt \quad (3-10)$$

To ensure the system's proper and stable dynamic response, the voltage loop bandwidth is set significantly lower than the current loop, making the current loop always capable of responding faster to any changes in i_d^* .

This alignment of voltage control is designed to regulate the angle between the rectifier and the generator, ensuring a stable output voltage magnitude regulation which directly influence the rectifier's reactive power output. In the small segment connection between generator and rectifier, the small signal model equation of the rectifier output can be seen as

$$V\Delta\alpha \approx X\Delta i_d \quad (3-11)$$

Where α represents difference or small deviation in the angle between the rectifier and the source generator with its voltage and angle evolve dynamically based on its internal electromotive force, and AVR action. However, for smaller electrical disturbances, the generator's terminal behaves sufficiently stiff for the same linearized voltage-current relationship to apply.

The rectifier voltage vector expressed within the generator's frame, can be written as:

$$V_d + jV_q = V \left(\cos(\theta_{gen} - \theta_{rect}) \right) + j \left(\sin(\theta_{gen} - \theta_{rect}) \right) \quad (3-12)$$

When the angle deviation/difference is small and the generator's terminal voltage magnitude varies slowly due to AVR dynamics, this reduces to

$$\theta_{gen} - \theta_{rect} \approx V_q \quad (3-13)$$

Assuming that both V_q and i_d are proportional to the angle difference or misalignment between the generator and converter's frame, serves as a means to control this angle of the system.

Thus, the output of this voltage controller can be defined using the small angle approximation. Which can be inferred that the rectifier is primarily controlled by two variables being the d-axis voltage (V_d), which is proportional to the voltage magnitude itself, and the q-axis current (i_q), which is proportional to the reactive power output.

$$1 - \frac{1}{2}(\theta_{gen} - \theta_{rect})^2 = V_d \quad (3-14)$$

The output of this voltage controller then becomes the primary input for the d-axis current regulator in the inner loop. No reactive power regulation is performed in this

system, thus the q-axis current reference remains at zero and as such, is fixed.

3.2.2.4 Current Control Loop (Inner Loop)

Reference current generated from the previous PI controller becomes the primary input for the inner dq-axis current control loop. The d-axis reference i_d^* represents the active-power demand required to maintain the DC-link voltage, while the q-axis reference i_q^* is fixed at zero because the system is designed for unity-power-factor operation. These references are compared with the measured dq currents, which are obtained by transforming the three-phase generator currents into the synchronous rotating reference frame using the same PLL angle that aligns the d-axis with the generator voltage.

From these references, the controller computes the instantaneous current errors:

$$e_d = i_d^* - i_d \quad (3-15)$$

$$e_q = i_q^* - i_q \quad (3-16)$$

Each error term is processed through a dedicated PI controller tuned for fast closed-loop response. The PI blocks generate preliminary voltage commands u_d and u_q , which represent the voltage levels required to drive the actual currents toward the commanded reference values.

$$u_d = K_{pc}e_d + K_{ic} \int e_d dt \quad (3-17)$$

$$u_q = K_{pc}e_q + K_{ic} \int e_q dt \quad (3-18)$$

At this stage, however, the dq current dynamics are still inherently coupled due to the rotating reference frame. Without compensation, changes in one axis would unintentionally influence the other axis. To eliminate this, decoupling feedforward terms are added. Specifically $+\omega L i_q$ is injected into the d-axis command, and $-\omega L i_d$ added into the q-axis command.

The natural coupling terms of the dq current dynamics:

$$v_d^* = u_d + \omega L i_q \quad (3-19)$$

$$v_q^* = u_q - \omega L i_d \quad (3-20)$$

After compensation terms are added:

$$v_d^* = u_d + \omega L i_q \quad (3-21)$$

$$v_q^* = u_q - \omega L i_d \quad (3-22)$$

These additional signals remove the effect of the synchronous rotation and make the d-axis and q-axis dynamics behave as two independent first-order control loops. This decoupling significantly improves accuracy during transients and ensures that the current loop can follow i_d^* and i_q^* without cross-axis distortion.

As stated in the voltage loop, designing the current and voltage loops requires a crucial step, for each loop designed to receive a different approach. Important things to mention is that each controller's bandwidth is always designed to have approximately 10 times the bandwidth of the previous controller. An example is the current control's bandwidth should have at least 10 times the value of the voltage control bandwidth, but also restricted to having the bandwidth limit's at a tenth of its rectifier / inverter's switching frequency. This study ensures that the defined bandwidth and design of the current controller are able to operate at a higher speed when compared to other control loops that are being implemented simultaneously. The current controller's bandwidth implemented in this model is 150 Hz which is permissible as it fulfills the limitation of 1 kHz (a tenth of $f_{sw} = 10kHz$). With the variables addressed, the gain of the plant can be calculated via the pole-zero cancellation as shown.

$$\begin{aligned} G_{loop}(s) &= \frac{i_g}{V_{generator}} \\ &= \frac{1}{sL + R} \end{aligned} \quad (3-23)$$

After compensation terms are added:

$$K_p^i = \omega_{bandwidth} L \quad (3-24)$$

$$\omega_{bandwidth} = 2\pi f_{bw} \quad (3-25)$$

$$K_i^i = \frac{R}{L} K_p^i \quad (3-26)$$

3.2.3 RL Filter Calculation

Determining the RL filter that is acting as the source impedance using the same procedure stated in the fundamental theory section, guarantees the filter design to perform a good stability by limiting the ripple injected to the generator stator from the rectifier, reducing the switching harmonics, lowering the Total Harmonic Distortion (THD) in a hybrid powertrain system.

Designing the RL filter starts with determining the cutoff frequency (f_c) of the filter. The cutoff frequency is chosen based on the degree of magnitude from the fundamental frequency of the system (50 Hz)

$$f_c = (2 - 4) \cdot f_{systemfundamental} = 150Hz \quad (3-27)$$

Choosing a very small damping ratio with the value of 0.1, ensures that the filter provides a good balance between transient response and overshoot. The inductance value of the RL filter is then calculated as follows.

$$L_{RL} = \frac{R}{2\pi \cdot f_c} = 1.06157 \times 10^{-4} \quad (3-28)$$

3.2.4 LCL Filter Calculation

Calculating the LCL filter using the same procedure stated in the fundamental theory section, guarantees the filter design to perform a good stability in a hybrid powertrain system.

$$\Delta I_{LCL,max} = 10\% I_b \sqrt{2} = 231.64818A \quad (3-29)$$

The maximum ripple is estimated from around 10% of the maximum base current, the value is then used along with the switching frequency and the DC-link's voltage as the basis to calculate the first inductor's inductance of L_{conv}

$$L_{conv} = \frac{1500VDC}{6 \cdot 10kHz \cdot 231.64818A} = 1.07922 \times 10^{-4}H \quad (3-30)$$

After gaining L_{conv} , the approximation of the filter's generator side inductor is estimated using the equation in and the capacitance is chosen by estimating the total reactive that flows at the base frequency into the capacitor. The value lies around 2% - 4%, in this case uses 3% for the system. Therefore the filter capacitance value is 0.03 times the value of its base capacitance

$$C_f = 0.03C_B = 0.03 \cdot 1.37285 \times 10^{-2} = 4.11855 \times 10^{-4}F \quad (3-31)$$

Following the capacitance of the LCL, the generator side inductor (L_g) provides the final stage of harmonic suppression necessary to satisfy the Total Harmonic Disturbance requirements imposed by the generator. L_g is calculated based on the required attenuation factor (A_{sw}) at the switching frequency (f_{sw}), ensuring that the harmonic remains are below the generator's limit.

$$L_{gen} = r \cdot L_{conv} = 1 \times 1.07922 \times 10^{-4}H \quad (3-32)$$

Damping resistor are calculated based on the damping ratio of the LCL's transfer function under the hybrid system's impedance scenario. The resonance frequency of the LCL filter can be determined as such:

$$Z_{LCL}(s) = sL_{gen} + \frac{1}{sC_f + \frac{1}{sL_2}} \quad (3-33)$$

$$\omega_{res}^{finite} = \sqrt{\frac{L_{conv} + L_{gen}}{L_{conv} \cdot L_{gen} \cdot C_f}} = 6707.91937 \quad (3-34)$$

Then the damping ratio (D_f) is chosen to be around $\frac{1}{6}$ or approximately 0.707 in order to provide a good balance between transient response and overshoot. The damping resistor value is then calculated as follows:

$$R_f = \frac{2D_f}{C_f \cdot \omega_{res}^{finite}} = 0.0120655 \quad (3-35)$$

3.2.5 RC Filter Calculation

Calculating the RC filter using the same procedure stated in the fundamental theory section, guarantees the filter design to a perform a good stability in the hybrid power-train system. This filter is place at two distinct and key position, namely the DC-Link and the distribution between the inverter and the induction motor each with their respective uses. The capacitance of the RC filter place in the DC link is determined based on the ripple current and the maximum allowable voltage ripple in the DC link.

$$I_{DC} = \frac{P_{nom}}{V_{DC}} = \frac{2MVA}{1500V} = 1333.33A \quad (3-36)$$

Using the value of the I_{DC} , with its estimated value of the base current in the DC link, along with the ripple frequency of the general system (50 Hz) and the maximum

allowable voltage ripple (ΔV_{max}) of 5% of the DC-link voltage, the capacitance value is calculated as follows:

$$C_{DC} = \frac{I_{DC}}{2 \cdot \omega_{ripple} \cdot \Delta V_{DC}} = 2.9622 \times 10^{-2} F \quad (3-37)$$

With $\omega_{ripple} = 2\pi f_{ripple}$ and $f_{ripple} = 6 \cdot 50Hz$ being the base frequency of the system multiplied by six due to the rectifier's operation. The damping resistor is calculated based on the desired damping ratio (D_{DC}) of the RC filter's transfer function. Based on the resonance of the angular frequency. The damping resistor can be calculated as shown in:

$$L_{DC} = \frac{V_{DC}}{\Delta I_{RC,DClink} \cdot 4 \cdot f_{sw}} = 2.8132 \times 10^{-5} H \quad (3-38)$$

From ??, in typical distribution system, the existence of L_{DC} is due to the parasitic inductance of the DC link cabling and connections.

$$R_d = 2 \cdot \frac{L_{DC}}{C_d} = 1.8994 \times 10^{-3} \quad (3-39)$$

While the RC filter placed in the DC-link between the rectifier and the inverter is calculated based on the ripple current and the maximum allowable voltage ripple in the DC link. The filter resides between the inverter and the induction motor to suppress high frequency harmonics produced by the inverter's switching action before reaching the motor windings using switching frequency of the inverter. Using the of the ΔI_{max} , with its estimated value of 10% of the base current, along with the switching frequency and the maximum allowable voltage ripple (ΔV_{max}) of the base voltage, and the inverter's inertial smoothing inductance between the phases. The capacitance value is calculated as follows:

$$C_f = \frac{\Delta I_{max}}{8 \cdot f_{sw} \cdot \Delta V_{max}} = 6.02206 \times 10^{-4} F \quad (3-40)$$

The damping resistor is calculated based on the desired damping ratio (D_f) of the RC filter's transfer function. Based on the resonance of the angular frequency and the damping ratio of 0.5 to 1 to achieve critical damping scenario. The damping resistor can be calculated as shown below:

$$L_f = \frac{V_{base}}{\Delta I_{max} \cdot 4 \cdot f_{sw}} = 1.03785 \times 10^{-4} H \quad (3-41)$$

$$\omega_{res} = \frac{1}{\sqrt{L_f \cdot C_f}} = 4000.57985 \text{ rad/s} \quad (3-42)$$

$$R_f \approx 2 \cdot D_f \cdot \sqrt{\frac{L_f}{C_f}} = 0.41514 \Omega \quad (3-43)$$

3.2.6 Determining Battery Model

In this system, the battery pack is configured as a pure series string, where each cell/module contributes additively to the total DC-link voltage while carrying the same current. A series topology is the only practical configuration for achieving the required DC-link voltage of 1500 V, because individual battery modules typically have nominal voltages in the range of 500 V per module. To meet the inverter and rectifier operating voltage derived from a 680 V line-line AC output and the power-processing requirement of a 2 MVA traction system the DC link must reach approximately 1500 V to provide modulation headroom, ensure stable operation under load, and maintain sufficient voltage margin for field-oriented control (FOC) of the induction motor.

$$I_{DC,nom} = \frac{P_{battery}}{V_{DC}} = \frac{500kW}{900V} = 333.33A \quad (3-44)$$

The full parameters of the series battery pack used in this thesis are summarized in the table below:

Parameter	Value
DC Link Voltage V_{dc}	1500 V
Rated System Power S	2 MVA
DC Current at Full Load I_{dc}	1333 A
Battery Module Nominal Voltage (typical)	500 V per module
Number of Series Modules	3 modules

Table 3.3. Series Battery Pack Parameters for the 1500 V DC-Link System

3.2.7 VVVF Inverter Parameters

The VVVF inverter used in this thesis is a three-phase voltage source inverter (VSI) that utilizes IGBTs as the primary switching devices. The inverter is designed to convert the DC voltage from the rectifier into a variable frequency and variable amplitude AC voltage to drive the induction motor. The specifications and parameters of the inverter are detailed below:

Parameter	Value
Power Rating	2 MVA
Nominal Frequency	50 Hz
Input DC Voltage	1500 V DC
Output Line-Line Voltage	680 V AC
Output Nominal Voltage (V_{rms})	400 V AC
Rated Current	1638 A
Switching Frequency	10 kHz

Table 3.4. VVVF Inverter Specifications

3.2.8 Induction Motor Specifications

Acting as the primary load and traction machine employed in this thesis, the proposed machine is a three-phase Squirrel Cade Induction Motor (IM). With the consideration of robust structure, low maintenance requirement, and its suitability for high performance variable speed applications when applied with the correct control method.

The specification and parameters that were used as model are detailed below:

Parameter	Value
Power Rating	373 kVA
Nominal Frequency	50 Hz
Line-Line Voltage	680 V AC
Nominal Voltage (V_{rms})	400 V AC

Table 3.5. Motor Specifications

Parameter	Value
Lm	0.0347 H
Lr	0.0355 H
Ls	0.0348 H
P	50 Hz
Tr	680 V AC
Ts	400 V AC
Llr	$8 \cdot 10^{-4}$ H
Lls	$1 \cdot 10^{-4}$ H
Rr	0.2280Ω
Rs	0.0870Ω

Table 3.6. Motor Specifications

These parameters will be crucial in controlling the system, with the primary target of controlling the Induction Motor's speed, while still maintaining torque and current levels.

3.2.9 Field Oriented Control

The Field control of the Induction system involves in transforming the variables of the outputed motor's value into a signal, specifically an SVPWM signal to control the inverter's current output. The FOC model that this system employs is a cascaded feedforward indirect rotor flux oriented control that utilizes the flux linkages and the currents in a feedback like matter. This type of control is chosen since this type of control only needed three primary variables namely the stator currents (i_s), the fluxes (ϕ) and the angular velocity of the motor (ω_m) produced by the motor shaft.

Procedure start at transforming the stators current (i_s), from the stationary ABC frame, into the rotational dq frame that aligned with the rotor's flux vector. Using both clark and park transformation.

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & \frac{-1}{2} & \frac{-1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}. \quad (3-45)$$

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}. \quad (3-46)$$

Where θ is the instantaneous angle of the rotating reference frame of the motor. In the indirect rotor-flux orientation θ is chosen so that the d-axis is aligned with the rotor flux vector (ϕ_r), therefore when $i_{rq} = 0$ in the rotor frame. This constraint allows the rotor flux magnitude to be regulated exclusively through the d-axis current (i_d), while solely using i_q to control the outputting torque.

In this thesis, the model implements a current-model rotor flux estimator, which uses the measured stator currents and the motor parameters L_m, L_r, R_r to reconstruct the rotor flux. The continuous rotor flux dynamic under simplifications is approximated by a first order system:

$$\phi_r(s) \approx \frac{L_m}{1 + sT_r} \cdot I_{sd}(s) \quad (3-47)$$

With the time constant of the rotor (T_r) being the division of the rotor inductance (L_r) with the rotor resistance (R_r). On the continuous terms, it behaves with a low pass behaviour, rotor flux follows stator d-current with time-constant (T_r). Converting the relation in discrete terms, using a discrete observer to integrates the (i_{sd}) with the rotor and stator's time constant yields the estimated (ϕ_r).

$$\phi_r(z) = \frac{L_m \left(\frac{T_s}{T_r} \right)}{z + \frac{(T_s - T_r)}{T_r}} \cdot I_{sd}(z) \quad (3-48)$$

The accurate estimation of the electrical angle (θ) is determined from two components, namely the rotor mechanical speed (ω_m) and the slip frequency derived from the electromagnetic torque and rotor flux dynamixs. The implementation used in this thesis follows the exact structure of the Simulink block shown previously, which combines model-based slip computation, discrete-time integration, and angle wrapping. Under the rotor-flux alignment, the angular frequency of the rotating reference frame is given by:

$$\omega_e = \omega_r + \omega_{sl} \quad (3-49)$$

$$\omega_r = \omega_m + \frac{P}{2} \quad (3-50)$$

Where ω_r is the rotor mechanical speed in electrical rad/s, and ω_{sl} is the slip frequency represents how fast can the stator magnetic field rotates relative to the rotor magnetic field. In rotor-flux-oriented FOC, these slip frequency is directly determined by the torque-producing current i_q and the estimated rotor flux ϕ_r .

$$\omega_{sl} = \frac{L_m}{T_r} \cdot \frac{i_{sq}}{\phi_r} \quad (3-51)$$

This equation arises from the induction machine torque expression under the rotor-flux orientation condition, where the torque is proportional to the product of i_{sq} and ϕ_r .

$$T_e = \frac{3}{2} \frac{P}{2} \frac{L_m}{L_r} \cdot i_{sq} \phi_r \quad (3-52)$$

Along with the fundamental flux dynamic equation,

$$\dot{\phi}_r = \frac{L_m}{T_r} i_{sd} - \frac{1}{T_r} \phi_r \quad (3-53)$$

Thus the slip frequency is tightly linked to the torque production mechanism of the induction motor. By controlling i_{sq} , the FOC scheme can directly influence the slip frequency, and consequently the rotor speed and torque output.

The synchronous angle (θ) is obtained through discrete-time integration of the electrical angular frequency (ω_e) with the wrapping operation to ensure it remains within the range of 0 to 2π radians.

$$\theta(k) = \theta(k-1) + \omega_e(k)T_s \quad (3-54)$$

$$\frac{KT_s}{z-1} \quad (3-55)$$

Implementing the sampling time of (T_s) and this discrete method avoid drifts, ensuring stability at high sampling rates, allowing for a precise coupling with the digital controllers. Added the properties of angle wrapping, the angle (θ) is always constrained within a single electrical revolution, this prevents angle overflow and maintains numerical stability in long-term operation.

The speed controller is implemented as a discrete PI controller that processes the speed error ($\omega^* - \omega_m$) to generate the torque-producing current reference (i_{sq}^*). The flux-producing current reference (i_{sd}^*) is set based on the desired flux level, often kept constant for maximum efficiency operation.

$$e_w(t) = \omega_m^*(t) - \omega_m(t) \quad (3-56)$$

$$T_e^*(t) = K_{pw}e_w(t) + K_{iw} \int e_w(t)dt \quad (3-57)$$

$$U[k] = U[k-1] + K_p(e[k] - e[k-1]) + K_i T_s e[k] \quad (3-58)$$

This torque command is then feeds into the torque-current conversion block, which uses the estimated rotor flux (ϕ_r) to compute the required i_{sq}^* . There is a key similarity that the rotor flux oriented FOC has with the VoC method, is that the outer loops must have a lower bandwidth than the inner loops. This is to ensure that the inner current loops can respond faster to changes in the reference values generated by the outer loops. In this thesis, the current controller follows the same rule as the rectifier's inner loop, having a bandwidth in the range from 100 to near $\frac{1}{10}f_{sw}$, while the speed controller is designed to have a bandwidth around 10 times lower than the current controller. Thus, using the same method as the rectifier's current controller, the gain of the plant can be calculated via the pole-zero cancellation as shown in (3-26).

3.2.10 Full System Configuration

The complete hybrid system equations are shown, the synchronous generator, rectifier, DC link, inverter, and induction motor. Each subsystem is modeled with its respective dynamics and control strategies, forming an integrated framework for analysis and simulation.

The differential equations that depicts the dynamic of this test system is shown throughout (3-59) to (3-81).

$$\frac{dV_{d,gen}}{dt} = -R_s I_{d,gen} + \omega_{gen} V_{q,gen} + \omega_{gen} L_q I_{q,gen} \quad (3-59)$$

$$\frac{dV_{q,gen}}{dt} = -R_s I_{q,gen} - \omega_{gen} V_{d,gen} - \omega_{gen} L_d I_{d,gen} + \omega_{gen} \lambda_f \quad (3-60)$$

$$\frac{dI_{d,gen}}{dt} = \frac{1}{L_d} (V_{d,gen} - R_s I_{d,gen} - \omega_{gen} L_q I_{q,gen}) \quad (3-61)$$

$$\frac{dI_{q,gen}}{dt} = \frac{1}{L_q} (V_{q,gen} - R_s I_{q,gen} + \omega_{gen} (L_d I_{d,gen} + \lambda_f)) \quad (3-62)$$

$$\frac{d\omega_{gen}}{dt} = \frac{1}{J} (\tau_m - \tau_e - B\omega_{gen}) \quad (3-63)$$

$$\frac{dI_{d,RL}}{dt} = \frac{1}{L_s} (V_{d,gen} - R_s I_{d,RL} + \omega L_s I_{q,RL}) \quad (3-64)$$

$$\frac{dI_{q,RL}}{dt} = \frac{1}{L_s} (V_{q,gen} - R_s I_{q,RL} - \omega L_s I_{d,RL}) \quad (3-65)$$

$$\frac{dI_{d1}}{dt} = \frac{1}{L_1} (V_{d,rect} - R_1 I_{d1} + \omega L_1 I_{q1} - V_{d,c}) \quad (3-66)$$

$$\frac{dI_{q1}}{dt} = \frac{1}{L_1} (V_{q,rect} - R_1 I_{q1} - \omega L_1 I_{d1} - V_{q,c}) \quad (3-67)$$

$$\begin{aligned} \frac{dV_{d,c}}{dt} &= \frac{1}{C_f} (I_{d1} - I_{d2}) + \omega V_{q,c} \\ \frac{dV_{q,c}}{dt} &= \frac{1}{C_f} (I_{q1} - I_{q2}) - \omega V_{d,c} \end{aligned} \quad (3-68)$$

$$\frac{dI_{d2}}{dt} = \frac{1}{L_2} (V_{d,c} - R_2 I_{d2} + \omega L_2 I_{q2} - V_{d,grid}) \quad (3-69)$$

$$\frac{dI_{q2}}{dt} = \frac{1}{L_2} (V_{q,c} - R_2 I_{q2} - \omega L_2 I_{d2} - V_{q,grid}) \quad (3-70)$$

$$\frac{d\theta_{gen}}{dt} = \omega_{gen} \quad (3-71)$$

$$\frac{d\omega_{gen}}{dt} = K_p e_q + K_i \int e_q dt \quad (3-72)$$

$$\frac{dI_{d,rect}}{dt} = \frac{1}{L_r} (V_{d,gen} - V_{d,dc} - R_r I_{d,rect} + \omega L_r I_{q,rect}) \quad (3-73)$$

$$\frac{dV_{dc}}{dt} = \frac{1}{C_{dc}} (I_{rect} - I_{inv} - \frac{V_{dc}}{R_{dc}}) \quad (3-74)$$

$$\frac{dI_{bat}}{dt} = \frac{1}{L_{bat}} (V_{dc} - R_{bat} I_{bat} - V_{oc}(SOC)) \quad (3-75)$$

$$\frac{dSOC}{dt} = -\frac{I_{bat}}{Q_{bat}} \quad (3-76)$$

$$\begin{aligned}\frac{dI_{d,inv}}{dt} &= \frac{1}{L_{inv}}(V_{d,inv} - V_{d,ac} - R_{inv}I_{d,inv} + \omega L_{inv}I_{q,inv}) \\ \frac{dI_{q,inv}}{dt} &= \frac{1}{L_{inv}}(V_{q,inv} - V_{q,ac} - R_{inv}I_{q,inv} - \omega L_{inv}I_{d,inv})\end{aligned}\quad (3-77)$$

$$\begin{aligned}\frac{dV_{d,ac}}{dt} &= \frac{1}{C_{ac}}(I_{d,inv} - \frac{V_{d,ac}}{R_{ac}}) + \omega V_{q,ac} \\ \frac{dV_{q,ac}}{dt} &= \frac{1}{C_{ac}}(I_{q,inv} - \frac{V_{q,ac}}{R_{ac}}) - \omega V_{d,ac}\end{aligned}\quad (3-78)$$

$$\begin{aligned}\frac{d\psi_d}{dt} &= V_{d,ac} - R_s I_d + \omega_s \psi_q \\ \frac{d\psi_q}{dt} &= V_{q,ac} - R_s I_q - \omega_s \psi_d\end{aligned}\quad (3-79)$$

$$\begin{aligned}I_d &= \frac{1}{L_s}(\psi_d - L_m I_{r,d}) \\ I_q &= \frac{1}{L_s}(\psi_q - L_m I_{r,q})\end{aligned}\quad (3-80)$$

$$\frac{d\omega_m}{dt} = \frac{1}{J}(\tau_e - \tau_L - B\omega_m) \quad (3-81)$$

All of these equations are transformed into terms of the frequency domain using the Laplace transform and serves as the basis for the small signal model of the entire hybrid system.

3.2.11 Small Signal Analysis

This section outlines the methodology employed to assess and derive the stability of the hybrid powertrain systems. Basing the approach on a linear system theory, which requires the linearization of the nonlinear differential equations that govern the system dynamics around a defined operating point.

3.2.11.1 Linearization: First Order Taylor Expansion

To begin analyzing a non-linear system, it is essential to linearize the system equations around a specific steady state operating point (x_0, u_0) where the system is in equilibrium would satisfy the terms $\dot{x}_0 = f(x_0, u_0) = 0$. This is typically achieved using a first-order Taylor series expansion by introducing a small perturbations $(\Delta x, \Delta u)$ into the states and the inputs. For a general nonlinear system described by the state-space equations:

$$f(x) \approx f(x_0) + \left. \frac{\partial f}{\partial x} \right|_{x_0} (x - x_0) + \left. \frac{\partial f}{\partial u} \right|_{u_0} (u - u_0) \quad (3-82)$$

This would lead to the creation of a Jacobian matrix which is structured for a n -state system ($x = [x_1, x_2, \dots, x_n]^T$) and can be arranged as shown in:

$$A = \left. \frac{\partial f}{\partial x} \right|_{(x_0, u_0)} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \quad (3-83)$$

$$B = \left. \frac{\partial f}{\partial u} \right|_{(x_0, u_0)} = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} & \dots & \frac{\partial f_1}{\partial u_m} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} & \dots & \frac{\partial f_2}{\partial u_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \frac{\partial f_n}{\partial u_2} & \dots & \frac{\partial f_n}{\partial u_m} \end{bmatrix} \quad (3-84)$$

Ignoring the second and higher order terms, the linearized state-space representation of the system can be expressed as shown below.

$$\begin{aligned} \dot{x} &= A\Delta x + B\Delta u \\ y &= C\Delta x + D\Delta u \end{aligned} \quad (3-85)$$

Where the state matrix (A) describing the system dynamics, input matrix (B) defining how inputs affect the states, output matrix (C) relating the states to the outputs, and feedthrough matrix (D).

3.2.11.2 Small-Signal Stability Analysis

Small-signal stability analysis is employed to evaluate the dynamic behavior of the proposed hybrid powertrain system when subjected to small disturbances around its nominal operating point. Although the system is fundamentally nonlinear, leaning to switching actions, dq-abc coordinate transformations, magnetic characteristics, and electromechanical coupling its local stability can be rigorously examined through linearization. By linearizing the complete nonlinear differential equations using a first-order Taylor expansion around the steady-state operating point, the entire model can be approximated as a linear state-space representation.

$$\Delta \dot{x} = A\Delta x + B\Delta u \quad (3-86)$$

Here, the Jacobian matrix A encapsulates the local dynamic interactions among electrical, mechanical, and control states. Eigenvalue analysis of A provides insight into oscillatory modes, damping ratios, and participation factors. This establishes whether the

system remains stable under slight perturbations that typically arise during train operation, such as minor fluctuations in generator current, small variations in DC-link voltage, or slight deviations in motor torque demand.

The purpose of this subsection is to verify that the integrated control architecture comprising generator-side voltage-oriented control (VoC), PLL synchronization, DC-link regulation, inverter-side field-oriented control (FOC), and motor electromechanics exhibits robust local stability across a broad range of realistic operating conditions.

To capture the influence of key subsystems and interfaces, several parameter-variation scenarios are evaluated:

- **DC-Link Parameter Variation:** This scenario examines the impact of DC-link passive elements on small-signal stability. Variations in $C_{dc} = 1 \times 10^{-2} - 2.9622 \times 10^{-2}$ and $R_{dc} = 0.1 \times 10^{-3} - 1.8994 \times 10^{-3}$ allow observation of how energy buffering and voltage damping characteristics alter the eigenvalue structure.
- **Generator-Side Control Parameter Variation (VoC Loop):** The stability of the generator rectifier is evaluated by varying d-axis and q-axis current controller gains, including the integrator gains $K_{id} = 1 - 5$, $K_{iq} = 1 - 5$ and proportional gains $K_{pd} = 0.750 - 1$, $K_{pq} = 0.750 - 1$ for the outer loop. While the integrator gains for the inner current loop ranges from $K_{id} = 50 - 200$, $K_{iq} = 50 - 200$ and proportional gains $K_{pd} = 25 - 100$, $K_{pq} = 25 - 100$ PLL bandwidth and damping are also modified to study their influence on voltage tracking and synchronization. The highest value for each of K_p and K_i attained from the aforementioned gain method.
- **Inverter-Side FOC Parameter Variation:** The inner current control loop gains and speed-loop gains are varied to investigate their effect on motor-side dynamic modes. Variations include inner loop controls $K_{pi}^{inv} = 50 - 200$, $K_{ii}^{inv} = 5 - 13$, and speed controller gains $K_{pw}^{inv} = 25 - 50$, $K_{iw}^{inv} = 13 - 26$. This analysis reveals how inverter current regulation and motor speed control interact to influence overall system stability, with the highest value for each of K_p and K_i attained from the aforementioned pole-zero cancellation method.

The outcomes of the small-signal stability investigation provide a detailed understanding of how each subsystem contributes to the global system stability. By observing eigenvalue migration under parameter variations, the analysis identifies critical oscillatory modes, evaluates damping performance, and validates the coherence of generator-side and motor-side control strategies. This establishes a rigorous foundation that the

proposed hybrid train architecture is dynamically stable and robust in its linearized neighborhood of operation.

3.2.11.3 Transient and Steady-State Analysis

Whereas small-signal analysis captures local stability characteristics, transient and steady-state analyses explore the full nonlinear dynamic performance of the hybrid train powertrain under realistic operating scenarios. These analyses reflect operational conditions such as sudden motor torque commands, fast generator rectifier transitions, battery charge/discharge events, and disturbances in AC-side voltage.

Transient simulation assesses the system's capability to withstand abrupt disturbances by observing time-domain responses of electrical and mechanical states. Steady-state analysis complements this by verifying that the system consistently settles to a stable operating condition after transients decay.

The scenarios considered in this section are as follows:

- **Harmonic in DC-Link's Voltage via Load:** A sudden load change is applied to the DC link to induce voltage harmonics. This scenario evaluates the rectifier's VoC response, DC-link voltage regulation, and inverter current behavior under non-ideal DC conditions. The load is modelled to be a simple resistor with varying values. Load step: $\Delta R_{load} = 1, 10, 50, 100, 500, 1000\Omega$
- **Speed Variation Test on the Induction Motor (FOC):** A step change is applied to the motor speed reference to evaluate torque response, current regulation, and disturbance rejection. Induction motor phase currents I_a, I_b, I_c are examined for oscillatory behavior. Speed step: $\Delta\omega_m = 550, 600, 750, 800, 900, 1000$.
- **Long-Term Steady-State Performance:** After disturbances fade, the system is evaluated for steady-state quality including:
 - DC-link voltage stability,
 - current ripple minimization,
 - motor speed convergence,
 - electromagnetic torque smoothness,
 - generator current balance and power flow consistency.

The transient and steady-state analyses collectively validate the real-world feasibility and operational robustness of the proposed hybrid train system. By testing its response to abrupt disturbances and verifying long-term stability, this section demonstrates that the control strategies VoC on the rectifier, FOC on the inverter, and DC-link coordination are capable of maintaining safe, stable, and efficient train operation. These

nonlinear simulation results complement the small-signal findings and confirm that the system performs satisfactorily in both dynamic and steady-state conditions.

3.2.12 Research Flow

The research flow of this thesis is structured to systematically develop, analyze, and validate the proposed hybrid diesel-electric powertrain system. The key stages of the research process are outlined below:

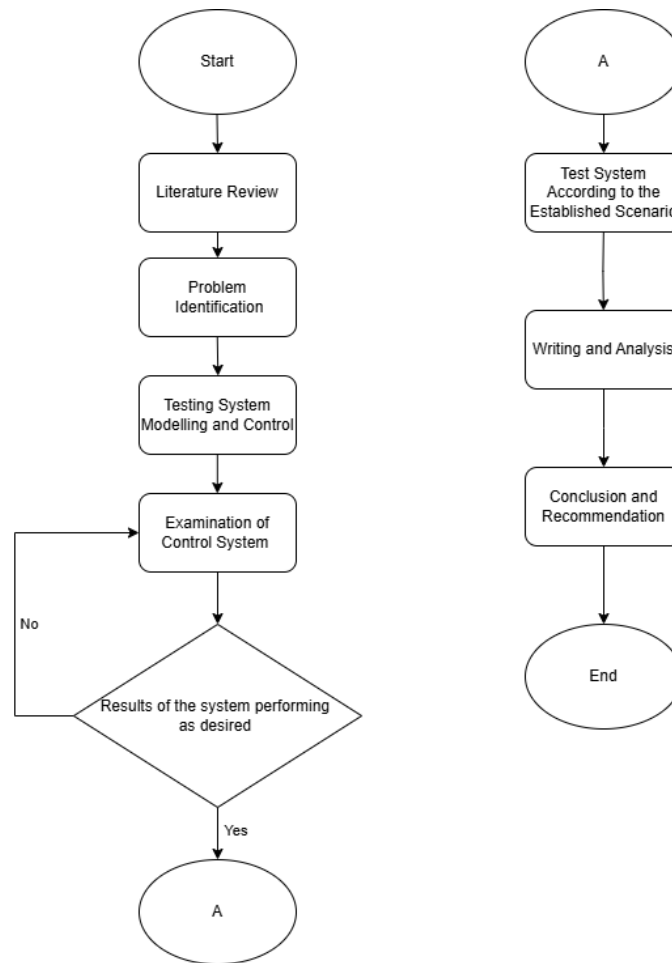


Figure 3.3. Research Flow Diagram

CHAPTER IV

RESULT AND DISCUSSION

4.1 Study Case 1: DC link Load Variation Impact on System Dynamics

In this section, the impact of varying DC link load on system dynamics is analyzed. The value of resistors that are used in this study case are 100 and 1000 Ω , to show the difference regarding the DC link under a relatively heavy load and a lighter load. The following figures illustrate the results obtained from simulations under different load conditions.

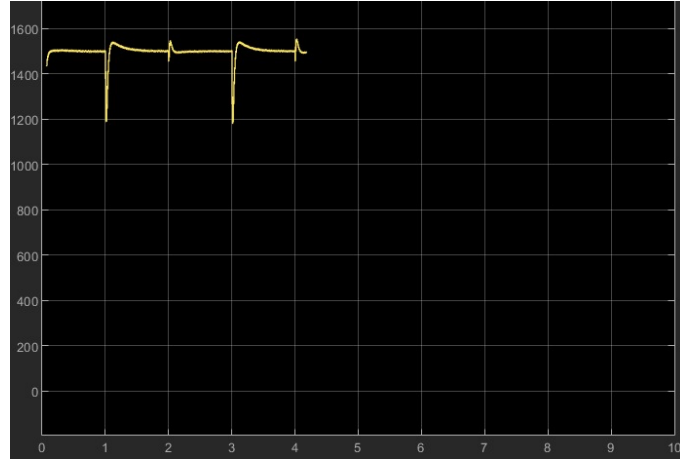


Figure 4.1. V_{dc} under 100 Ω load

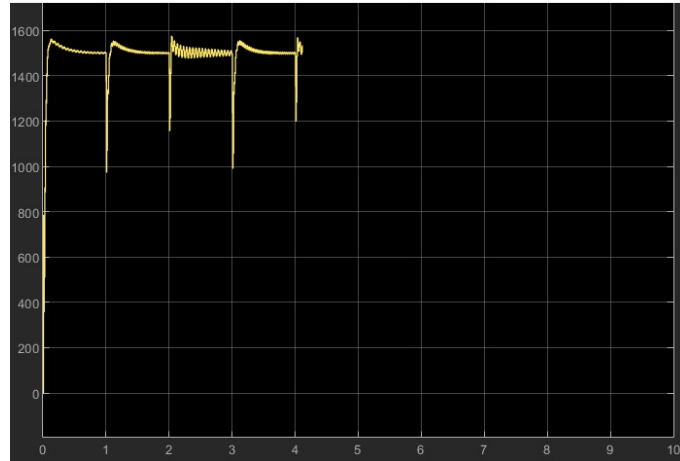


Figure 4.2. V_{dc} under 1000 Ω load

Although a 1000 Ω resistance represents a lighter electrical load compared to 100 Ω , the experimental results show that the DC-link voltage becomes less stable under the lighter loading condition. This behavior arises from the dynamic characteristics of the transformer rectifier capacitor system used in the setup.

Under the $1000\ \Omega$ load, the transformer operates far below its rated loading level. When lightly loaded, the secondary voltage tends to rise above its nominal value, and the waveform becomes more susceptible to distortion and high-frequency components. At the same time, the DC-link capacitor discharges very slowly because only a small amount of current is drawn by the high-resistance load. As a consequence, the capacitor remains close to its peak voltage for most of the cycle, and the diodes conduct only during narrow intervals around the AC voltage peaks. These narrow conduction intervals produce sharp, spike-like charging pulses that introduce noticeable harmonic content and irregularities into the DC-link voltage. The result is a DC waveform that, although high in magnitude, exhibits increased ripple sharpness and a visibly less stable profile.

In contrast, the $100\ \Omega$ load draws significantly more current, causing the transformer to operate closer to its intended loading conditions. The secondary voltage therefore becomes more regular and less prone to distortion. Because the heavier load causes the DC-link capacitor to discharge more substantially between cycles, the diodes conduct over a wider portion of each AC period. These wider conduction intervals create smoother and more evenly distributed charging pulses, which in turn produce a DC-link voltage that, while exhibiting somewhat larger amplitude ripple, maintains a more uniform and stable waveform shape. The overall appearance is one of improved robustness, with fewer harmonic distortions and fewer abrupt variations.

Thus, despite the heavier electrical demand imposed by the $100\ \Omega$ resistor, the corresponding DC-link voltage appears more stable and better regulated. Causing a very light loading conditions reduce natural damping and increase sensitivity to harmonic disturbances, while moderate or heavy loading tends to stabilize the voltage waveform and suppress high-frequency irregularities.

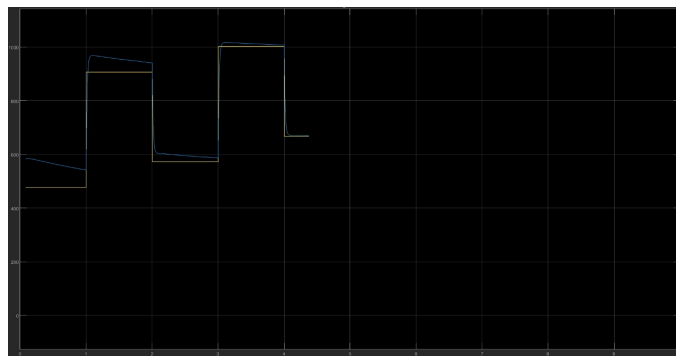


Figure 4.3. Motor angular velocity influenced by V_{dc} under $100\ \Omega$ load

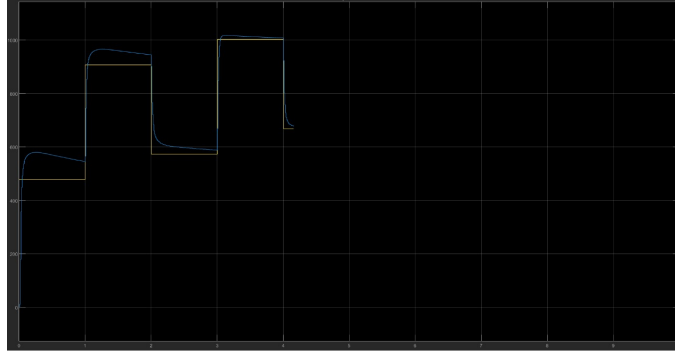


Figure 4.4. Motor angular velocity influenced by V_{dc} under $1000\ \Omega$ load

The motor's angular velocity closely follows the dynamic variations of the DC-link voltage, and this relationship becomes evident when comparing the responses under $100\ \Omega$ and $1000\ \Omega$ load conditions. Because the motor drive depends directly on the stability and magnitude of the DC-link voltage, any disturbance or fluctuation in the DC-link immediately influences the torque production and mechanical speed of the motor.

Under the $1000\ \Omega$ loading condition, the DC-link exhibits more irregular and harmonic-rich behavior due to the lightly loaded rectifier and the narrow, spike-like charging pulses produced at the peaks of the AC waveform. These voltage irregularities are transferred to the inverter output, generating small but frequent variations in the motor's input voltage. As a result, the motor experiences slight but continuous torque oscillations that manifest as noticeable ripple in the angular velocity. Although the overall speed remains close to its nominal operating point, the waveform of the angular velocity contains higher-frequency fluctuations and appears less stable. This outcome reflects the sensitivity of the motor drive to even small disturbances in the supply, especially when the DC-link lacks the damping effect provided by a heavier load.

In contrast, the motor's behavior under the $100\ \Omega$ load is significantly more stable. With the DC-link voltage exhibiting smoother and more uniform ripple, the inverter receives a steadier power supply, allowing the motor to develop torque in a more consistent manner. The broader conduction intervals in the rectifier produce a more predictable voltage profile, which translates into reduced torque ripple and a smoother mechanical response. Although the heavier load causes the DC-link to experience deeper but more uniform voltage sag, these variations occur gradually and systematically, enabling the motor control system to respond more effectively. Consequently, the angular velocity waveform under the $100\ \Omega$ load appears firmer, cleaner, and more robust, with fewer high-frequency disturbances.

Overall, the comparison demonstrates that motor speed stability is influenced not only by the magnitude of the load but also by the quality and harmonic content of the DC-link voltage. The lighter $1000\ \Omega$ load leads to a less stable angular velocity due

to the presence of harmonic-rich voltage fluctuations, whereas the heavier $100\ \Omega$ load results in a smoother and more stable mechanical response because of the more uniform and better-damped DC-link voltage waveform.

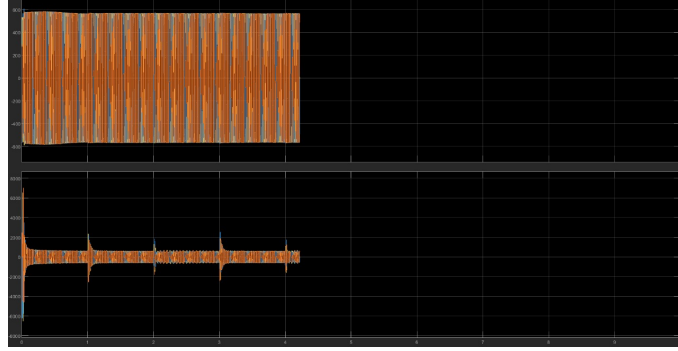


Figure 4.5. V_t and I_t of generator under load

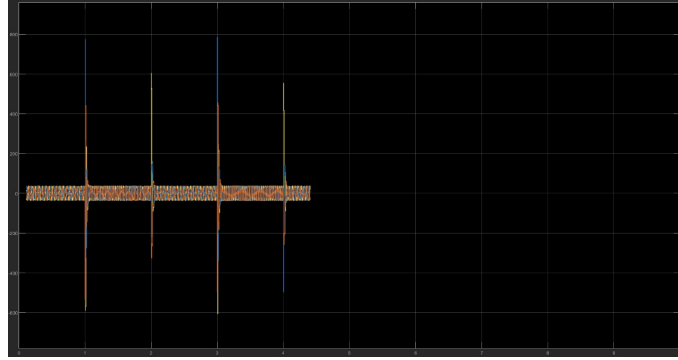


Figure 4.6. Inverter current under DC link load of 100 ohm

The behavior of the generator under a $100\ \Omega$ load reflects the increased electrical demand imposed on the system and the dynamic interaction between the generator, the rectifier, and the DC-link capacitor. As the $100\ \Omega$ load draws significantly more power from the DC link, the generator is required to deliver a higher and more sustained output current in order to maintain the DC-link voltage and supply the motor drive. This demand is clearly visible in the generator's terminal voltage (V_t) and current (I_t) waveforms.

When the $100\ \Omega$ load is applied, the generator terminal current increases sharply as the system adjusts to the new power requirement. This rise in current is accompanied by transient fluctuations that occur as the generator and rectifier settle into a new equilibrium of power delivery. The generator's excitation system responds to these changes by regulating the terminal voltage, which may momentarily dip or oscillate before stabilizing. These small voltage variations reflect the generator's internal electromagnetic dynamics and the rectifier's non-linear loading characteristics.

Once the transient effects subside, the generator operates with a higher and more consistent current output, supplying the necessary power to the rectifier and DC-link. Al-

though the terminal voltage may decrease slightly compared to lighter loading conditions, the generator maintains a stable voltage profile appropriate for the heavier load. The waveform generally appears smoother and more uniform compared to a lightly loaded condition, because the generator is being driven closer to its intended operational region, where its magnetic flux, torque production, and excitation control are more effectively balanced.

The generator current waveform under the $100\ \Omega$ load also exhibits periodic ripples caused by the rectifier's conduction pattern. As the rectifier draws current in broader intervals due to the capacitor discharging more substantially under the heavier load the generator delivers current in a more continuous and predictable manner. This results in current waveforms that, while containing the expected rectification ripple, show a steady overall magnitude that corresponds directly to the higher power output.

In summary, the generator voltage and current under the $100\ \Omega$ load reflect a system delivering substantial power with stable and predictable electrical characteristics. The generator operates in a more controlled region of its performance curve, producing smoother voltage and current waveforms and supplying the DC link with the increased power demanded by the heavier load.

4.2 Study Case 2: DC link Load Variation Impact on System Dynamics without LCL filter

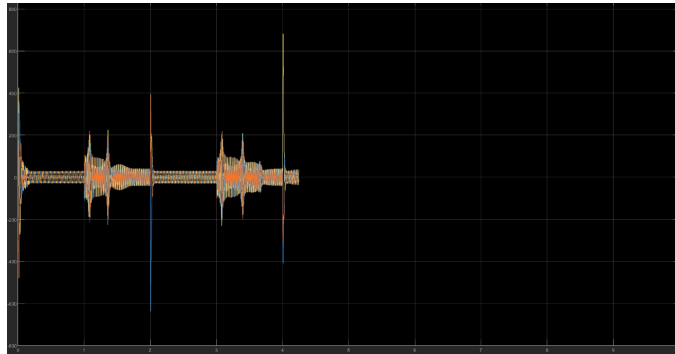


Figure 4.7. Inverter current under DC link load of $100\ \Omega$ without LCL filter

The inverter output current provides the clearest evidence of the consequences of omitting an LCL filter. In a well-filtered system, the LCL network presents high impedance at the switching frequency while allowing the fundamental component to pass with minimal distortion. Without this buffer, the inverter's PWM switching waveform appears almost unattenuated in the output current. As a result, the measured current contains substantial high-frequency components, visible as pronounced ripple and fast-varying distortions superimposed on the fundamental waveform.

This unfiltered current not only represents poor harmonic performance but also contributes to undesirable interactions within the rest of the system. The switching harmonics propagate into the load and back to the DC link, reinforcing the oscillatory phenomena observed in V_{dc} . In a practical implementation, such current distortion would significantly increase switching losses, EMI emissions, and thermal stress in both the inverter switches and the load. Therefore, the inverter current waveform qualitatively confirms that the absence of an LCL filter directly degrades current quality and increases harmonic energy throughout the system.

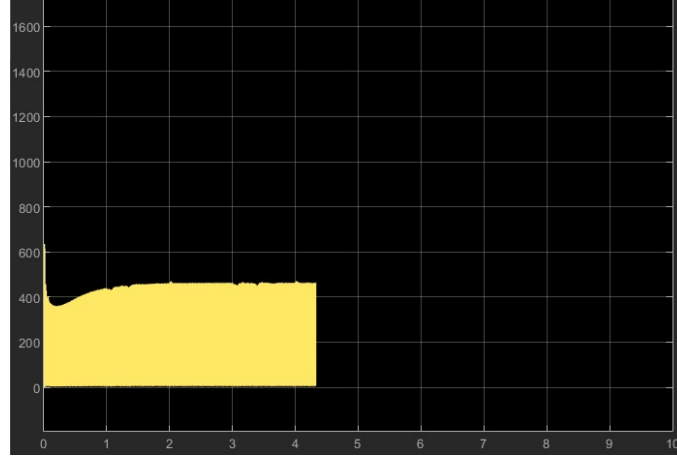


Figure 4.8. V_{dc} under DC link load of $100\ \Omega$ without LCL filter

The DC-link voltage response under a $100\ \Omega$ load shows that the system reaches its nominal operating point; however, the steady-state waveform contains a noticeable band of oscillation that persists throughout the entire simulation interval. This behavior is a direct manifestation of the system operating without an LCL filter, which normally provides attenuation of switching harmonics and decouples the DC link from inverter-side disturbances.

In the absence of such filtering, the high-frequency switching current generated by the inverter is injected directly into the DC-link capacitor. Because the capacitor now becomes the sole element responsible for absorbing the instantaneous power ripple, its voltage is modulated by these fluctuations, creating the oscillatory pattern visible in the results. The oscillations remain bounded, indicating that the system is not unstable; however, the damping characteristics are clearly insufficient. The control loops do not fully suppress the switching-induced ripple, and the natural LC resonance formed by the DC-link capacitor and stray inductances is continuously excited. Consequently, V_{dc} exhibits marginal damping, representing degraded dynamic performance that would otherwise be mitigated by an LCL network.

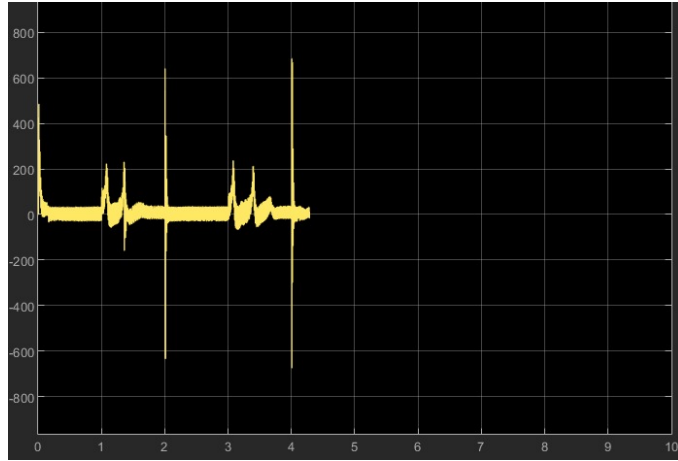


Figure 4.9. I_{dc} under DC link load of $100\ \Omega$ without LCL filter

Input currents drawn from the rectifier or DC source exhibits a smooth average trajectory, but with fluctuations that correlate strongly with the DC-link ripple. These fluctuations correspond to variations in instantaneous input power and are more pronounced precisely because no LCL filter exists to shield the DC side from inverter-generated disturbances. In a filtered configuration, the LCL network would confine most of the switching ripple to the AC-side interface, leaving the DC input current comparatively clean.

Here, the rectifier is directly exposed to the inverter's switching harmonics. As V_{dc} oscillates due to inadequate filtering and marginal damping, I_{dc} adjusts accordingly, producing a dynamic profile that is measurably noisier than would be expected from a system equipped with an LCL filter. Although the current remains within stable bounds, the quality of the input current is degraded. This behavior would increase DC-source stress and could negatively influence upstream components, validating the importance of filter isolation in reducing ripple transmission.

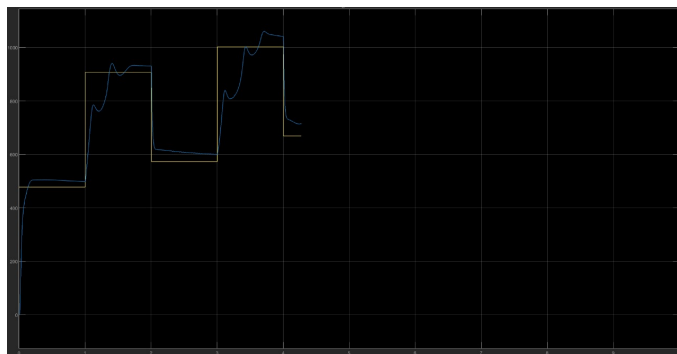


Figure 4.10. Motor angular velocity influenced by V_{dc} under $100\ \Omega$ load without LCL filter

The mechanical speed response demonstrates that the speed-control loop and the mechanical inertia provide effective low-frequency damping, resulting in a stable steady-

state speed. However, the waveform also exhibits small deviations that originate from electrical torque ripple produced by distorted phase currents.

In the absence of an LCL filter, the motor receives current with a much higher harmonic content. Since electromagnetic torque is directly proportional to the product of the current and the back-EMF, any harmonic distortion in the current causes torque pulsations at corresponding frequencies. These torque pulsations, though filtered by the mechanical inertia, still introduce slight speed ripple observable in the rotor speed waveform.

In a practical drive system, prolonged exposure to such torque ripple can lead to increased mechanical loss, resonance excitation in flexible couplings, and reduced overall servo performance. The observed rotor-speed behavior therefore confirms not only electrical consequences of removing the LCL filter but also the mechanical implications of passing unfiltered switching harmonics into the motor.

CHAPTER V

CONCLUSION AND RECOMMENDATION

5.1 Conclusion

This thesis has presented the design, modeling, and stability analysis of a state-feedback-based control system for a DC-link coupled hybrid electric locomotive (HEL) powertrain. The primary objective was to address the critical challenge of maintaining DC-link voltage stability under highly dynamic load conditions while coordinating power flow between a diesel-driven synchronous generator, a directly coupled battery energy storage system (BESS), and a traction induction motor.

A comprehensive nonlinear state-space model with 26 state variables was developed in the MATLAB/Simulink environment, capturing the electromechanical dynamics of the complete system, including the governor-engine, self-excited synchronous generator (SESG), LCL filter, bidirectional PWM rectifier, 1500 V DC-link, battery pack, VVVF inverter, and induction motor with Field-Oriented Control (FOC). The model was systematically linearized to facilitate small-signal stability analysis and controller design.

The core of the control strategy involved the implementation of a cascaded control architecture: Voltage Oriented Control (VoC) for the generator-side PWM rectifier to regulate the DC-link voltage and Field-Oriented Control (FOC) for the inverter to provide precise torque and speed control of the traction motor. The simulation results from two primary study cases lead to the following conclusions:

1. The integrated control architecture successfully maintains system stability. The proposed VoC and FOC strategies, combined with properly designed passive filters (RL, LCL, RC), effectively coordinated the power flow. The system demonstrated stable operation under both heavy ($100\ \Omega$) and light ($1000\ \Omega$) DC-link resistive loads, with all key states—DC-link voltage, generator currents, and motor speed—reaching and maintaining their target operating points.
2. DC-link voltage quality and motor performance are intrinsically linked to loading conditions and filtering. Study Case 1 revealed that a heavier load ($100\ \Omega$) resulted in a more stable and uniform DC-link voltage waveform due to broader rectifier conduction angles and better damping. Consequently, the motor's angular velocity response was smoother and more robust. In contrast, a lighter load ($1000\ \Omega$) led to a higher but less stable DC-link voltage with increased harmonic content, which translated into noticeable speed ripple in the motor.
3. The LCL filter is a critical component for harmonic attenuation and system damping. Study Case 2, which simulated the system without the LCL filter, provided

clear evidence of its importance. The absence of the filter led to:

- Pronounced high-frequency switching ripple in the inverter output current.
 - Sustained, marginally-damped oscillations in the DC-link voltage (V_{dc}).
 - Increased harmonic distortion in the DC input current (I_{dc}).
 - Small but observable torque and speed ripple in the motor due to unfiltered current harmonics. These results confirm that the LCL filter is essential for decoupling the switching harmonics of the power converters from the DC-link and the source, thereby ensuring current quality, reducing component stress, and enhancing overall system stability.
4. The state-feedback framework provides a viable foundation for advanced control. The linearized state-space model and the accompanying small-signal analysis establish a rigorous foundation for designing and tuning state-feedback controllers. This approach paves the way for future work to directly regulate all system states for optimal dynamic performance and robustness.

In summary, this research demonstrates that a well-designed control system employing VoC and FOC, supported by appropriate passive filtering, can effectively manage the complex power interactions in a DC-link coupled hybrid train. The system maintains stability across varying loads, validating the proposed architecture as a promising solution for enhancing the efficiency and reliability of modern hybrid railway propulsion systems.

5.2 Recommendation

While this thesis provides a comprehensive simulation-based analysis, several avenues exist for extending and improving upon this work:

1. Implementation of Full State-Feedback Control: The immediate next step is to synthesize and implement an optimal state-feedback controller based on the linearized model developed in Chapter III. A comparative analysis between the cascaded PI-based control (VoC/FOC) and the state-feedback controller should be conducted to quantify improvements in transient response, disturbance rejection, and robustness.
2. Development of an Energy Management System (EMS): This work focused on the electrical dynamics and stability. A supervisory Energy Management System (EMS) should be developed to intelligently coordinate power flow based on real-time demands. This includes optimizing the split between diesel generation and battery discharge/charge, managing State of Charge (SoC), and implementing regenerative braking strategies to maximize fuel efficiency and battery life.

3. **Inclusion of Detailed Battery and Loss Models:** Future models should incorporate more sophisticated, temperature-dependent battery models that account for aging effects. Furthermore, non-ideal switching losses (conduction and switching) in the semiconductor devices (IGBTs/MOSFETs) and copper losses in machines should be included for a more accurate efficiency analysis and thermal design.
4. **Experimental Validation:** The most critical recommendation is to validate the simulation findings through a scaled-down laboratory prototype. An experimental testbed with a lower-power synchronous generator, battery bank, and induction motor drive would allow for the practical verification of control algorithms, filter performance, and stability under real-world disturbances and parameter variations.
5. **Investigation of Advanced Topologies and Fault Scenarios:** Research could explore the use of advanced converter topologies (e.g., three-level Neutral Point Clamped converters) for higher efficiency and reduced harmonic distortion. Additionally, the system's response and control strategies under fault conditions (e.g., DC-link short circuit, battery disconnect, motor phase loss) should be analyzed to ensure operational safety and reliability.

By addressing these recommendations, the research on hybrid locomotive power-trains can progress from theoretical modeling and simulation toward practical, optimized, and resilient systems ready for real-world deployment.

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